Who Benefits from Benefits?
Municipal Consequences of Federal Cash Transfers

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Abstract

Centralized welfare programs are designed to limit local discretion, yet their implementation often depends on municipal bureaucracies. We study whether local politicians can obtain electoral returns from a federally designed cash transfer when the remaining discretion lies in registration. In Brazil's *Bolsa Família*, eligibility rules, benefit values, and payments are set nationally, while municipal welfare offices register eligible households. We exploit a 2009 reform that changed the formula assigning municipal beneficiary quotas, generating differential authorized expansion across municipalities for reasons unrelated to local politics. In the 2,105 municipalities where the 2008 incumbent mayor sought reelection in 2012, the reform increased realized enrollment and improved incumbent performance. A one-standard-deviation quota shock raised incumbent vote share by about one percentage point in reduced form; instrumenting enrollment with the quota shock, an additional 1,000 beneficiary families raised vote share by approximately one percentage point. The return is conditional on local implementation capacity. The identifying first stage and electoral response are concentrated in municipalities with a pre-existing *CRAS* welfare office and, within those municipalities, where the incoming mayor replaced the inherited front-line manager. The shock did not cause manager turnover or new *CRAS* openings. Nor do substitute managers appear to be partisan loyalists: they are not selected through prior affiliation with the mayor's party or coalition, and they do not subsequently realign with it. Instead, they later run for local office at roughly twice the rate of stayers, consistent with the managerial post becoming a platform for local political careers. Centralized rules and payment flows therefore do not depoliticize transfers whose last mile remains locally administered; program design travels only as far as local capacity carries it.

Keywords: cash transfers; electoral returns; state capacity; *Bolsa Família*.
JEL Codes: D72, H53, H75, I38, O17.

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1 Introduction

Across the developing world, cash transfer programs have become a leading tool for reducing poverty while insulating redistribution from local political discretion. By standardizing eligibility rules, fixing benefit values by federal decree, and processing payments through centralized banking systems, these programs are designed to reduce the scope for patronage and discretionary allocation (Fiszbein and Schady, 2009). Yet access often depends on local bureaucracies controlled by elected officials. Whether control over program registration rather than program design is sufficient to generate electoral returns is the question we address.

We study this question in the context of Brazil's *Bolsa Família*, one of the largest conditional cash transfer programs in the world.[1] Although eligibility rules, benefit size, and payments are determined by the federal government, municipalities remain responsible for identifying and registering eligible households, mostly through local welfare centers, the CRAS (*Centro de Referência de Assistência Social*). This institutional arrangement creates a narrow but important margin of local discretion: while mayors cannot alter transfer amounts, redirect funds, or withhold payments, they control the administrative pipeline through which households may gain access to the program.

This margin depended on a registration process that required active municipal effort. Identifying potentially eligible families meant locating them, conducting household interviews, and collecting documentation; municipal authorities controlled the local registration apparatus, which gave them influence over the pace, intensity, and reach of enrollment efforts.[2] In practice, two municipalities facing similar levels of underlying need could generate very different realized coverage depending on how actively the local administration pursued registration.

Our empirical strategy exploits a 2009 reform that revised the formula used to compute municipal *Bolsa Família* quotas. The program operates through federally determined municipal quotas, which set a ceiling on the number of families that can be served in each municipality.[3] The 2009 reform updated the formula used to calculate

[1] By June 2006, *Bolsa Família* covered 11.1 million families, approximately 46 million people, or about 25% of the Brazilian population. By June 2015, the program supported 13.8 million families, with an average monthly benefit of R\$167.15 and a budget equivalent to 0.5% of GDP.

[2] Hellmann (2015) describes enrollment as a municipal responsibility. He reports that as of September 2014, 77% of registrations were made without a home visit. Municipalities were responsible for preparing the local structure for registration, training personnel, promoting awareness campaigns, and designing *active search* strategies to reach eligible families outside the existing registry.

[3] Lindert et al. (2007) explains that *Bolsa Família* maintained municipal program quotas based on local poverty estimates. Original municipal allocations were constructed by combining the 2001 PNAD with the 2000 Census, generating an initial national target of 11.2 million families when the program was

those municipal ceilings using census and survey information, generating large and heterogeneous changes in the number of families each municipality was authorized to serve.[4] Crucially, the reform expanded the ceiling, not the registry: although already-registered eligible families were absorbed automatically, the magnitude of the expansion left a large residual margin of unfilled slots whose realization depended on local governments' effort to identify, locate, and register eligible families not yet in the federal database.

We combine administrative data on quotas, realized beneficiary counts, and local social-assistance infrastructure with municipal electoral data from the subsequent election. To isolate the variation due solely to the formula change, we recenter the quota shock against a counterfactual quota constructed from the pre-reform allocation rule (Borusyak and Hull, 2023) and use it to instrument realized beneficiary expansion, estimating its effect on incumbent electoral performance. This is the empirical counterpart to the institutional fact that federal authorization required local action to materialize.

Our findings trace the reform along four stages: the first-stage relationship between federal authorization and realized enrollment; the intention-to-treat effect of the authorization shock on electoral outcomes; the average causal response of electoral outcomes to realized enrollment that combines the two; and the institutional conditions under which these effects emerge.

The first stage confirms that federal authorization traveled to realized enrollment through the channel the institutional design implies. A one-standard-deviation increase in the quota shock, roughly 843 additional authorized slots, expanded realized beneficiary coverage by 0.198 standard deviations, approximately 980 additional families.

Once federal authorization is shown to translate into realized enrollment, we can estimate the electoral return to expanding the ceiling itself, before separating out how much of that ceiling municipalities chose to fill. This intention-to-treat estimate is 0.99 percentage points on the incumbent's vote share: the electoral consequence of authorizing more beneficiaries, integrating over the local take-up response documented in the first stage.

Scaling the intention-to-treat effect by the first stage gives the average causal re-

created in 2003.

[4] Hellmann (2015) notes that *Bolsa Família* administered municipal quotas using estimates of poor families developed in partnership with IBGE and reports that, in 2009, the federal government refined the measurement of the target population by incorporating methodologies to capture income volatility among vulnerable households. These studies formed a major expansion of coverage, incorporating more than 1.6 million additional families.

sponse of electoral outcomes to realized beneficiary expansion. In standardized terms, a one-standard-deviation increase in instrumented enrollment, about 4,949 additional families, raises vote share by 4.99 percentage points. The estimate is robust to alternative instrument thresholds, different measurement windows for the endogenous variable, other outcome definitions, distinct weighting schemes, and inverse-probability weighting for selection into the rerunning sample.

These returns travel through the institutional layer that converts authorization into delivery. In the 690 municipalities without a pre-existing CRAS in 2008, the quota shock generates a first stage roughly a fifth as strong as in municipalities with a CRAS, and the electoral effect cannot be identified at conventional precision for the no-CRAS subgroup. The estimator's implied weights concentrates approximately nine-tenths of the headline identifying mass on the 1,415 municipalities with a CRAS, where the local implementation chain was in place.

Within CRAS municipalities, a second binding margin opens around the front-line manager who runs the unit day-to-day. The expansion travels almost entirely through municipalities in which the incoming mayor replaced the inherited director after the 2008 election: where the manager was not replaced, the first stage attenuates sharply and the electoral effect cannot be pinned down at conventional precision. Three quarters of these replacements bring in individuals new to the municipal welfare bureaucracy rather than internal reshuffles, and the substitutes the shock attracts treat the post differently from inherited managers. These replacement managers do not appear to be partisan loyalists selected into office. Although many had prior party affiliations, they were no more likely than outgoing or retained managers to be affiliated with the mayor's party, nor did the shock lead them to join the mayor's party or coalition. Their distinctive response is electoral: they later run for local office at roughly twice the rate of the other groups.

These findings advance four related literatures. We speak first to the political economy of cash transfer programs. A large body of work shows that incumbents benefit electorally from the expansion of social transfers (Manacorda et al., 2011; De La O, 2013). Much of this literature studies settings in which local discretion is deliberately minimized, relying on centrally determined thresholds, randomized assignment, or other exogenous variation to identify electoral effects. We complement this work by studying a setting in which the program architecture is centralized but the last mile of delivery is not. Our findings suggest that centralizing transfer rules and payment flows is not sufficient to depoliticize redistribution when access to the program depends on a locally controlled registration bureaucracy. Even when mayors cannot alter eligibility criteria or benefit levels, control over the administrative pipeline through which households gain program access can still generate measurable electoral returns.

Our paper is especially closely related to Frey (2019), but the two designs identify electoral returns under different implementation regimes: Frey's intervention leaves limited room for mayoral involvement, whereas ours depends directly on municipal administration. Frey exploits a discontinuity in federal funding for the *Family Health Program*, a primary-care program in which teams (a doctor, a nurse, an assistant nurse, and community health agents) follow a federally specified protocol of routine home visits to poor households. His mechanism runs through an information channel embedded in those visits: as part of the protocol, the teams disseminate *Bolsa Família* eligibility information, and new enrollment follows mechanically, without any discretionary action by the municipal executive. We exploit the complementary margin. In our setting, converting expanded federal authorization into realized enrollment requires active effort from the municipal welfare staff that the mayor hires, deploys, and directs to identify, locate, and register eligible families; the channel is the local administration itself. The two designs also identify on nearly opposite parts of the municipal distribution: Frey's threshold design is concentrated among small, low-development municipalities near a funding threshold, whereas the 2009 quota reform expanded capacity primarily in more populous municipalities that had not yet absorbed earlier increases and still had room to add beneficiaries. Within his sample, coverage growth driven by an outreach channel that does not require mayoral effort reduces incumbency advantage and shifts the composition of challengers toward less clientelistic parties. Within ours, coverage growth driven by a channel that requires mayoral effort produces positive returns for the incumbent.

Second, we contribute to the literature on political accountability under shared governance. A central question in this literature is whether voters can correctly attribute responsibility when multiple levels of government jointly produce policy outcomes (Przeworski et al., 1999; Besley, 2006). Ferraz and Finan (2008) show that voters punish corrupt mayors when randomly assigned audits reveal misuse of federal funds, especially where local media facilitate information transmission. Avis et al. (2018) show that audits also reduce corruption through a legal disciplining channel. Our setting differs because the relevant information concerns positive delivery rather than malfeasance. Voters observe the expansion of a valued program and update their beliefs about the incumbent's competence and effort. The mechanism we document is therefore closer to electoral credit attribution than to sanctioning, but it operates through the same logic: voters use observable policy outcomes to infer politician type. In that sense, our results speak to a broader literature on attribution in multi-level settings by showing that voters can reward local incumbents even when the policy is federally funded and formally centralized, so long as local effort remains salient in implementation.

Third, we contribute to the literature on state capacity and the bureaucratic foundations of policy delivery. Besley and Persson (2009, 2011) establish state capacity as a precondition for credible policy delivery, and Finan et al. (2017) survey the personnel and bureaucratic infrastructure through which public programs are implemented. Our findings identify an extensive margin of local administrative capacity within this framework: whether the municipality had a CRAS in place before the expansion. Municipalities without a pre-existing CRAS converted essentially none of their quota shock into additional beneficiaries, while municipalities with a CRAS translated the shock into meaningful program expansion. The electoral consequences follow this same threshold: federal policy generates local political returns only where administrative infrastructure exists to carry it into implementation.

Fourth, conditional on the presence of a delivery apparatus, our within-CRAS findings on bureaucratic turnover speak to the literature on the selection and politicization of frontline bureaucrats. Manager turnover itself is not caused by the federal shock; it is election-driven, set by the 2008 mayoral transition that takes office months before the 2009 reform. Among municipalities where this turnover had already occurred, the headline estimate is concentrated in cases where the incoming mayor replaced the inherited frontline manager. Yet the substitutes those mayors install do not appear to be selected for prior partisan alignment. Two complementary mechanisms are consistent with what these substitutes do once installed, and the data do not adjudicate between them. First, program expansion raises the *de facto* attractiveness of the managerial post and pulls a more ambitious pool of candidates into it, consistent with Dal Bó et al. (2013). Second, the post itself becomes a launching pad: substitutes run for elected office at higher rates than stayers but do not switch to the mayor's coalition party, echoing Colonnelli et al. (2020), in which public-sector positions serve as channels into electoral careers, with the qualification that here the candidacy response is downstream of the post rather than upstream selection on partisan ties. The pattern departs from the loyalist-allocation equilibrium in Robinson and Verdier (2013) and from the hiring-discretion turnover in Akhtari et al. (2022): in our setting the turnover is election-driven, and politicization happens downstream rather than at hiring.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting of *Bolsa Família*, the quota system, the 2009 reform, and the data sources. Section 3 develops the conceptual framework: an augmented model of the federal-to-local implementation chain that isolates two binding margins for delivery, the presence of local welfare infrastructure and the alignment of front-line bureaucrats. Section 4 presents the empirical strategy, defines the instrument, and discusses the identifying assumptions. Section 5 reports the main results, the robustness checks, and the heterogeneity decomposition. Section 6 tests the proposed mechanisms and documents how the shock changes the selection and subsequent careers of substitute managers. Section 7 concludes.

2 Background and Data

2.1 Institutional Background

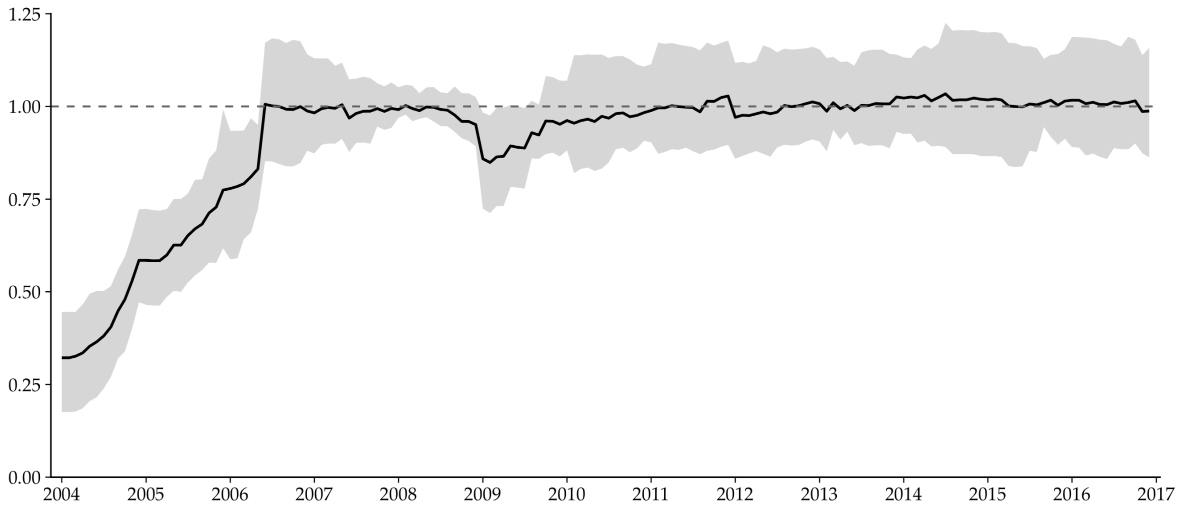
Launched in 2004, *Bolsa Família* consolidated several pre-existing social assistance programs into a single conditional cash transfer system, becoming Brazil's flagship poverty alleviation policy. In 2009, the program classified eligible families into two income groups. Families with monthly per capita income at or below R$70 were classified as extremely poor and received a basic monthly benefit of R$68, plus variable benefits of R$22 per child or adolescent aged 0–15, up to three, and R$33 per adolescent aged 16–17, up to two. Families with monthly per capita income above R$70 and at or below R$140 qualified only for the child- and adolescent-linked variable benefits. These benefits were conditional on age-specific education and health requirements. The maximum monthly transfer per family was R$200. By December 2009, the program reached approximately 12.4 million families, with an average benefit of about R$95.

Central to the program's operations is the *Cadastro Único* (*CadÚnico*), a national registry of low-income households that serves as the gateway to *Bolsa Família* and dozens of other social policies. The registry targets families with per capita income below half the minimum wage (R$232.5 in 2009), a threshold substantially higher than the program's own eligibility lines, so that the pool of registered families considerably exceeds the set of actual beneficiaries.

Eligibility criteria are nationally uniform, benefit values are fixed by federal decree, and payments are processed and deposited directly into beneficiaries' accounts by the state-owned bank *Caixa Econômica Federal*. Mayors cannot alter transfer amounts, redirect funds, or withhold payments. What municipal governments control is the registration process itself.

Registration in the *Cadastro Único* requires a face-to-face interview: a social worker

Figure 1: Realized Enrollment as a Share of Authorized Quota



Note: For each municipality-month, I first compute the ratio of realized *Bolsa Família* beneficiaries to the authorized quota. The solid line plots the quota-weighted cross-municipal mean of these municipality-level ratios. The shaded band spans the weighted 25th to 75th percentile. The dashed horizontal line marks full take-up (ratio = 1). Before 2007, municipalities operated well below their authorized ceilings, and the rapid convergence toward 1 reflects the administrative build-up of the program in its first years. After this initial convergence, realized enrollment generally tracked authorized capacity closely. The 2009 reform temporarily broke that alignment: expanded quotas opened a gap between authorized slots and realized beneficiaries, creating the local implementation margin through which municipalities had to identify, register, and enroll additional eligible families.

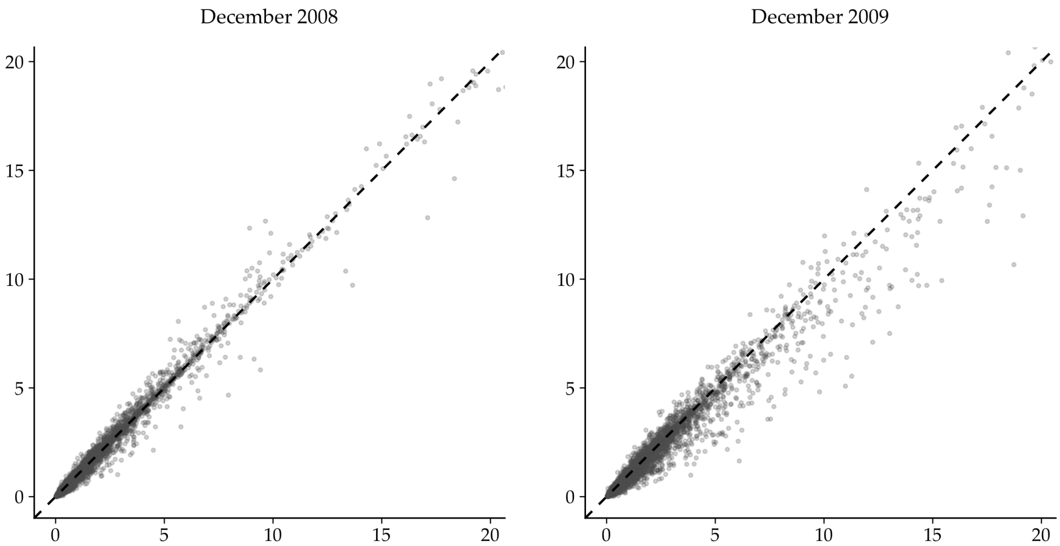
fills out a standardized questionnaire covering household composition, housing conditions, assets, employment, and income. In urban areas, families typically present themselves at a *Centro de Referência de Assistência Social* (*CRAS*). In rural and remote areas, however, eligible families cannot realistically be expected to travel to a registration point. There, coverage depends on active outreach: municipal teams must physically locate potential beneficiaries, travel to communities with limited road access, no telephone coverage, and no internet, and conduct household interviews on site. Federal regulations encourage this practice by recommending that interviews take place at the family's home, both to facilitate verification and to reach populations that would otherwise remain invisible to the registry.[5]

Eligibility alone does not guarantee benefit receipt. Access to *Bolsa Família* depends on the availability of municipal quotas: caps on the number of beneficiaries per municipality, defined nationally and revised approximately every three years based on poverty estimates derived from census and survey data. These quotas function as official upper bounds on municipal enrollment, but the constraint is not always imme-

[5]In 2009, there were approximately 5,800 *CRAS* units distributed across roughly 4,000 of Brazil's 5,568 municipalities.

diately binding. In practice, municipalities frequently operate near their ceiling, oscillating modestly above or below it from month to month before the federal priority mechanism reallocates unused slots.

Figure 2: Authorized Quota vs. Realized Enrollment

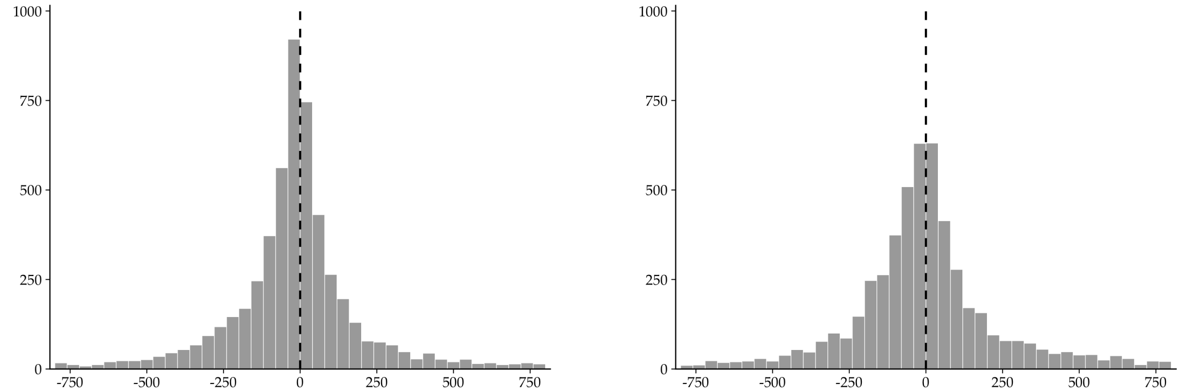


Note. Each dot is a municipality. The horizontal axis shows the federally authorized quota and the vertical axis shows the number of families actually enrolled in *Bolsa Família*, both in thousands. The dashed line marks the 45-degree locus of full take-up (realized = authorized). In December 2008, most municipalities lie close to or slightly below the 45-degree line, indicating that authorizations were approximately binding. In December 2009, following the federal quota reform, a substantial mass of municipalities shifts to the right of the 45-degree line, reflecting the expansion of authorized quotas that had not yet been translated into realized enrollment. The gap between the 45-degree line and the scatter in 2009 is the slack that our instrument exploits. Both axes are capped at the 99th percentile of the cross-municipal distribution (approximately 20,000 families); the small number of large urban municipalities above this threshold are excluded from the frame but not from any analysis.

The priority index considers both quota utilization and unmet registered demand: municipalities with low beneficiary-to-quota ratios and few eligible-but-unserved families risk seeing unused slots reallocated to places with stronger registered demand. The quota system therefore creates a use-it-or-lose-it implementation pressure. A quota increase is not a benefit guarantee, but it opens federally authorized room that municipalities must fill by identifying, registering, and updating eligible households. Effective coverage thus depends on local administrative initiative: political will and organizational capacity determine how much of the authorized ceiling becomes actual benefit receipt.

In 2009, the federal government revised the methodology used to compute municipal quotas, producing a heterogeneous reallocation of authorized beneficiary slots across municipalities. The previous approach, in place since 2006, distributed slots in

Figure 3: Distribution of Slack Between Authorized Quota and Realized Enrollment



Note. Each panel shows the distribution of slack, defined as the authorized quota minus realized enrollment, across municipalities in December 2009, immediately following the federal reform. The left panel uses the official authorized quota; the right panel uses the counterfactual quota. Positive values indicate that a municipality received an authorization it had not yet translated into realized enrollment. The dashed vertical line marks zero slack. Both distributions are trimmed at the 5th and 95th percentiles to limit the influence of extreme observations. The mass of positive values in both panels confirms that the reform expanded quotas beyond contemporaneous enrollment levels for a substantial share of municipalities.

proportion to each municipality's estimated number of poor families from the 2000 Census, adjusted for intercensal population growth. The 2009 revised methodology accounts for income volatility among families near the eligibility line, a feature of transient poverty that earlier projections systematically undercount (Gerard et al., 2025).

Alongside the methodological update, all municipal quotas were scaled by a factor of 1.18 to account for structural income volatility near the poverty threshold. Because this multiplicative factor is constant, it contributes no cross-municipal variation to the instrument; all identifying variation in the quota shock derives from the updated municipal poverty estimates and the change in income thresholds. Together, these changes added nearly 1.6 million new authorized slots in 2009. The reform expanded the ceiling, not the registry: although already-registered eligible families were absorbed automatically, the magnitude of the expansion left a large residual margin of unfilled slots whose realization depended on the municipal government's effort to identify, locate, and register eligible families not yet in the federal database.

2.2 Data Sources

Our empirical analysis combines four municipal-level data sources: administrative records on *Bolsa Família*, electoral returns, census and household-survey data, and administrative records on local social-assistance infrastructure.

We begin with monthly administrative records from the *Ministério do Desenvolvimento Social* on *Bolsa Família* quotas, realized coverage, and total transfers from 2006 to 2012. These data span the pre-reform baseline, the 2009 reallocation, and the post-reform expansion window through the 2012 election, and allow us to measure both authorized and realized program expansion.

We then use electoral records from the *Tribunal Superior Eleitoral* (*TSE*) for the 2008 and 2012 municipal elections. These data provide candidate-level vote totals, turnout, and detailed candidate characteristics, including education, prior occupation, and party affiliation.

To reconstruct counterfactual municipal quotas under the pre-reform allocation rule, we combine the 2000 Census from the *Instituto Brasileiro de Geografia e Estatística* (*IBGE*) with *Pesquisa Nacional por Amostra de Domicílios* (*PNAD*)-based estimates. The two census waves let us construct alternative versions of the baseline controls and approximate the quota formula that would have prevailed absent the reform.

Finally, we use administrative records from the *Censo SUAS* covering the universe of *CRAS* units in 2008 and 2009. In addition to information on the identity and tenure of directors and staff, these data include measures of local infrastructure and service provision, such as whether units conduct interviews and active searches.

2.3 Descriptive Statistics

The sample consists of 2,105 Brazilian municipalities observed in the 2008 and 2012 elections, those where the 2008 incumbent sought reelection. Table 1 reports descriptive statistics for the full population of municipalities and for this regression sample.

Three features stand out. The 2008 winner's vote share averages 56.3 percent in the full population and 52.9 percent in the regression sample; the tighter dispersion reflects the rerunning restriction, which selects against incumbents at the extremes of the 2008 outcome distribution. *Bolsa Família* enrollment and the quota shock are both heavily right-skewed: in 2009 the average municipality has 1,936 beneficiary families against a median of only 947, so both enter the regressions standardized. Census 2000 covariates have nearly identical means across panels, indicating the rerunning restriction does not select on the baselines we control for.

Table 1: Descriptive Statistics

	N	Min	Mean	Median	Max	SD
Panel A: All Municipalities						
Incumbent Won (2012)	4,730	0.00	55.35	100.00	100.00	49.72
Winner Vote Share (2008) (pp)	11,008	20.44	56.26	53.95	100.00	13.98
Realized Additional BF Expansion (2009)	11,008	-9,845.50	146.41	25.25	34,402.17	889.31
Excess Additional Beneficiaries (2009)	11,008	-20,173.15	8.39	23.03	58,544.62	1,250.69
Population (2000)	11,008	795.00	30,061.56	10,313.00	10,277,518.00	181,976.90
Gini (2000) (%)	11,008	30.00	54.73	55.00	87.00	6.82
Poor Population (2000) (%)	11,008	5.17	63.96	65.77	99.00	20.69
Income per capita (2000) (R$)	11,008	62.65	338.02	308.11	1,759.76	192.21
Formal Workers Population (2000) (%)	11,008	1.92	36.07	34.44	86.38	18.13
Urban Population (2000) (%)	11,008	0.00	58.88	59.53	100.00	23.35
Panel B: Regression Sample						
Incumbent Won (2012)	4,210	0.00	56.34	100.00	100.00	49.60
Winner Vote Share (2008) (pp)	4,210	24.06	52.92	52.69	99.16	9.41
Realized Additional BF Expansion (2009)	4,210	-889.25	145.41	27.83	34,402.17	947.24
Excess Additional Beneficiaries (2009)	4,210	-11,005.21	3.12	21.79	13,065.79	843.14
Population (2000)	4,210	795.00	27,876.69	10,556.00	5,773,130.00	145,264.20
Gini (2000) (%)	4,210	30.00	54.84	55.00	80.00	6.73
Poor Population (2000) (%)	4,210	7.66	64.10	66.78	99.00	20.97
Income per capita (2000) (R$)	4,210	69.20	337.21	300.12	1,759.76	195.03
Formal Workers Population (2000) (%)	4,210	2.89	35.78	33.99	86.38	18.28
Urban Population (2000) (%)	4,210	0.00	58.77	59.20	100.00	23.30

Notes: Panel A covers the universe of municipalities with a 2008 winner vote share and complete Census 2000 baseline controls; Panel B is the regression sample, municipalities where the 2008 incumbent sought reelection, subject to the data availability. Realized Additional BF Expansion is the muni-level inter-election change in the twelve-month moving average of beneficiary families between 2008 and 2012. Excess Additional Beneficiaries is the difference between the 2009 federal quota and the counterfactual quota, the quota that would have prevailed under the pre-reform allocation rule. The large standard deviation of this variable relative to its mean reflects substantial right skew driven by large urban municipalities; the regressions use a standardized version. The SD is smaller in Panel B than in Panel A because the regression sample drops municipalities without an incumbent rerunner. Full distributional statistics are reported in Appendix B.3.

3 Conceptual Framework

This section develops a simple framework for the political returns to welfare expansion. We proceed in three steps. The benchmark is a standard political-agency model in which local capacity is summarized by a single continuous implementation parameter: some municipalities can translate federal authorization into enrollment more easily than others, but all face the same basic technology. The augmented model then decomposes capacity into two institutionally grounded margins, the presence of an operating local delivery apparatus (*CRAS*, staff, and registration routines) and the the incoming administration's decision to replace the inherited front-line manager with its own appointee. These margins combine multiplicatively, so the first stage is concentrated where both are present. We close with a voter-response block that delivers the linear vote-share form and a partisan-attribution extension under which the inference channel sharpens when the mayor's party also operates the federal program. The framework gives a parsimonious structure for interpreting the heterogeneity exercises, the partisan asymmetry, and the mechanism analysis that follows.

3.1 Agents

The model has three sets of agents: a federal government, a continuum of municipalities indexed by m, each governed by an incumbent mayor, and a continuum of voters in each municipality indexed by i.

Federal government. The federal government is non-strategic. It sets a beneficiary quota Q_m^t for each municipality m in period t, computed from a national allocation rule based on poverty estimates and population growth. From the municipality's perspective, Q_m^t is exogenous: it determines the number of beneficiary slots the federal government authorizes locally, but not how many eligible families the municipality actually enrolls.

Mayor. At $t = 0$, the incumbent mayor in municipality m inherits a pre-treatment implementation environment: the local administrative apparatus available to register eligible families, the personnel operating that apparatus, and the routines through which information is transmitted to the federal registry. In the benchmark model, this environment is summarized by a scalar ψ_m, the share of the quota that can be filled through inherited capacity alone. In the augmented model, the same environment is decomposed into two margins, κ_m^I and κ_m^A, which capture the presence of an operating *CRAS* apparatus and the replacement of the inherited front-line manager by the incoming administration.

Conditional on this pre-treatment environment, the mayor chooses discretionary implementation effort $a_m \geq 0$: directing the welfare office to conduct active search, reallocating staff time toward registration, prioritizing documentation and data updates, and pressing the local apparatus to convert authorized slots into eligible families entered in the federal registry. The marginal cost of this effort depends on the mayor's unobserved type $\theta_m \in \{0, 1\}$, which represents administrative competence in the political-agency tradition surveyed by Ashworth (2012):

$$c(\theta_m) = c_0 - c_1 \cdot \theta_m, \qquad c_0 > c_1 > 0. \tag{1}$$

Type is drawn by nature, observed by the mayor, and unobserved by voters; competent mayors thus face a lower marginal cost of effort and choose strictly higher a_m in equilibrium.

Voters. Voters respond both to receiving the transfer and to what local implementation reveals about the incumbent.[6] The material channel reflects the documented response of beneficiaries to direct programmatic transfers in cash-transfer settings (Manacorda et al., 2011; Zucco, 2013; Avis et al., 2018). The inference channel follows models of political agency in which voters use observed government performance to infer an incumbent's unobserved competence (Ferejohn, 1986; Maskin and Tirole, 2004; Ashworth, 2012). Voter i's utility differential from supporting the incumbent versus the challenger is

$$\Delta U_{im} = \alpha\, d_{im} + \delta\ \Pr(\theta_m = 1 \mid s_{im}) + \eta_{im}, \tag{2}$$

where $d_{im} \in \{0, 1\}$ indicates whether household i in municipality m is a beneficiary, s_{im} is a noisy signal of program implementation, $\alpha > 0$ is the material reward to receiving the transfer, $\delta > 0$ is the value voters place on competent incumbents, and η_{im} is an idiosyncratic ideology shock.

Aggregate vote share. Aggregating individual choices across the continuum of voters under probabilistic voting in the tradition of Lindbeck and Weibull (1987), with the idiosyncratic shocks η_{im} smoothing across the ideology distribution, the municipality-level vote share for the incumbent takes the linear form

$$V_m = V_0 + \lambda_1 B_m + \lambda_2 e_m, \tag{3}$$

where $\lambda_1 > 0$ aggregates the material channel, with each beneficiary reducing resistance to supporting the incumbent through α in equation (2), and $\lambda_2 \geq 0$ aggregates the inference channel, with visible implementation raising the posterior $\Pr(\theta_m = 1)$ for all voters, beneficiaries and non-beneficiaries alike.

[6] The baseline model is agnostic about voter sophistication in separating federal from local contributions. The local component of delivery is directly observable: voters see *CRAS* operating, and registrations being processed locally. Whether voters additionally separate the federal component is immaterial for the electoral effect under the baseline: at most, a less sophisticated voter would also misattribute federal contributions, but the locally observable activity remains the proximate trigger. The partisan-attribution extension developed below loosens this agnosticism for the special case in which the local mayor's party brand matches the federal program's brand, allowing voters to use the partisan label as an additional, noisy attribution device.

3.2 Benchmark Model

In the benchmark, inherited implementation capacity is summarized by a scalar $\psi_m \in [0, \infty)$, the inherited capacity that translates into effort-free quota utilization. Realized program coverage is

$$e_m = \psi_m + a_m, \qquad B_m = e_m \cdot Q_m, \tag{4}$$

where e_m is the quota utilization rate and B_m is the number of realized beneficiaries.

The mayor's objective combines the electoral payoff, the convex cost of discretionary effort, and an institutional compliance pressure that penalizes deviations from full quota utilization, reflecting the federal monitoring of municipal performance against the authorized ceiling:

$$U_m = V_m - \tfrac{1}{2}c(\theta_m)a_m^2 - \tfrac{1}{2}\tau(e_m - 1)^2, \tag{5}$$

where V_m is the municipality-level vote share from equation (3) and $\tau > 0$ parameterizes the enforcement pressure to remain near the quota benchmark.

Proposition 1 (Optimal Effort and Realized Expansion)**.** *In the benchmark model, the unique optimal discretionary effort is*

$$a_m^*(\theta_m) = \frac{(\lambda_1 Q_m + \lambda_2) + \tau(1 - \psi_m)}{c(\theta_m) + \tau}. \tag{6}$$

Equilibrium realized coverage and beneficiaries are

$$e_m^*(\theta_m) = \psi_m + a_m^*(\theta_m), \qquad B_m^*(\theta_m) = e_m^*(\theta_m) \cdot Q_m. \tag{7}$$

The first stage of the empirical design satisfies

$$\frac{\partial B_m^*}{\partial Q_m} = e_m^*(\theta_m) + Q_m \cdot \frac{\lambda_1}{c(\theta_m) + \tau} > 0 \tag{8}$$

for all municipalities with $\psi_m \geq 0$, $Q_m > 0$, and $a_m^* > 0$.

Proof. Substituting equation (4) into the mayor's objective and differentiating with respect to a_m yields the first-order condition. Solving this condition gives equation (6); substituting the result back into $e_m = \psi_m + a_m$ and $B_m = e_m Q_m$ gives the remaining expressions. Appendix F provides the full derivation. □

The benchmark model predicts that the federal quota shock expands realized enrollment in every municipality, with the magnitude of expansion varying smoothly with inherited capacity ψ_m, quota size Q_m, and the mayor's administrative type θ_m. Heterogeneity in the first stage is therefore a matter of degree rather than of kind: the partial derivative $\partial B_m^*/\partial Q_m$ is strictly positive in every municipality with $\psi_m \geq 0$, $Q_m > 0$, and $a_m^* > 0$, so within-subgroup first stages cannot collapse to zero. The benchmark cannot accommodate a partition in which one subgroup's first stage approaches zero while another remains strongly identified, since $\partial B_m^*/\partial Q_m$ remains bounded below by the positive effort-response term $Q_m \cdot \lambda_1/[c(\theta_m) + \tau]$ even at the corner $\psi_m = a_m^* = 0$. Because realized enrollment enters vote share positively through the material channel, $\lambda_1 > 0$ in equation (3), the same logic implies a positive headline coefficient on incumbent vote share.

3.3 Augmented Model

The benchmark is useful precisely because it is restrictive: it treats local capacity as a single continuous implementation parameter. In this setting, however, administrative capacity is more modular. The personnel-economics-of-the-state literature emphasizes that public output depends not only on the existence of local administrative infrastructure, but also on the incentives and control of the front-line agents who operate it (Finan et al., 2017; Akhtari et al., 2022; Colonnelli et al., 2020). We therefore enrich the benchmark by decomposing capacity into two institutionally grounded margins.

Infrastructure component. We write the first capacity component as $\kappa_m^I \in \{0, 1\}$ indicating whether municipality m had a *Centro de Referência de Assistência Social* (*CRAS*) in 2008. The *CRAS* is a ready-made bureaucratic apparatus for enrolling households into social programs, with permanent staff, standardized protocols, and a direct data pipeline into the *Cadastro Único*. Finan et al. (2017) place this kind of pre-existing local infrastructure at the heart of the state-capacity literature: front-line administrative output is shaped by the agencies and personnel a municipality has on hand to execute federal mandates. Municipalities without a *CRAS* can still enroll new beneficiaries through alternative channels (public hospitals, mobile *cadastro* teams, ad hoc active outreach), but the productivity of those channels is lower than what an existing *CRAS* delivers. The same federal authorization therefore translates into less realized expansion where the apparatus is missing, and more where it is in place.

Alignment component. We write the second capacity component as $\kappa_m^A \in \{0, 1\}$ indicating whether the *CRAS* manager (the *coordenador* of the unit) was replaced after the 2008 municipal election, when the newly elected mayor took office. Two independent literatures motivate the relevance of front-line manager replacement as a productivity margin. Akhtari et al. (2022) document that mayoral transitions in Brazil systematically trigger turnover of street-level bureaucrats, with replacements concentrated in positions where the mayor has hiring discretion, including managers of social-assistance and education facilities. Colonnelli et al. (2020) document, in the same Brazilian setting, that public-sector positions function as electoral stepping stones, with their attractiveness as career-launching pads tied to the operational footprint of the office. Both readings imply that a post-2008 manager change can shift the productivity of the implementation chain: under the first, through replacement with bureaucrats more responsive to the incoming mayoral coalition, and under the second, through replacement with politically ambitious agents who exert more effort because program expansion raises the visibility of the post and its attractiveness as a launching pad for a local electoral career. The model treats κ_m^A as a generic productivity margin consistent with both readings.

Production with multiplicative capacity. The implementation chain that translates federal authorization into realized expansion has two indispensable capacity components: an existing registration apparatus and an active front-line manager. Kremer (1993) and Jones (2011) characterize technologies of this form: the productivity of the chain is the product of component qualities, so the failure of any single component governs the productivity of the whole.

We adopt this structure with a productivity floor $\gamma \in [0, 1)$ to allow attenuation rather than literal collapse when one capacity component is weak. Let

$$g(\kappa_m^I, \kappa_m^A) = \gamma + (1-\gamma)\, \kappa_m^I\, \kappa_m^A, \qquad \gamma \in [0,1), \tag{9}$$

so that $g = 1$ when both capacity components are active and $g = \gamma$ otherwise. Realized coverage takes the form

$$e_m = \psi_0 + g(\kappa_m^I, \kappa_m^A) \cdot (\bar{\psi} + a_m), \qquad B_m = e_m \cdot Q_m, \tag{10}$$

where ψ_0 and $\bar{\psi}$ are the floor and ceiling defined above. The strict Leontief case is the limit $\gamma = 0$; with $\gamma > 0$, subgroups that lack one capacity component still exercise some quotas, just at the attenuated rate g.

The mayor's objective remains as in equation (5), with V_m now driven by the augmented e_m and B_m from equation (10). Solving yields the following characterization.

Proposition 2 (Optimal Effort and Attenuated First Stage)**.** *In the augmented model with* *production* (10) *and multiplier* (9)*, optimal discretionary effort* $a^*_m(\theta_m)$ *is strictly increasing in the multiplier* $g(\kappa^I_m, \kappa^A_m)$*. The first stage satisfies*

$$\frac{\partial B^*_m}{\partial Q_m} = e^*_m(\theta_m) + Q_m \cdot \frac{\lambda_1 \, g^2}{c(\theta_m) + \tau \, g^2}, \tag{11}$$

which is strictly increasing in g*. As* $\kappa^I_m \kappa^A_m \to 0$ *the first stage attenuates by a factor of order* γ^2 *relative to the active subgroup. The attenuation operates on the marginal-productivity term and propagates through equilibrium effort.*

Proof. Substituting equation (10) into the mayor's objective and differentiating with respect to a_m yields the first-order condition with the multiplier $g(\kappa^I_m, \kappa^A_m)$ entering both the marginal electoral return to effort and the marginal compliance term. Solving this condition gives optimal effort as an increasing function of g. Differentiating equilibrium beneficiaries $B^*_m = e^*_m Q_m$ with respect to Q_m then yields equation (11). Appendix F provides the full derivation. □

The static-response component of equation (11) attenuates linearly in γ through $g\bar{\psi}$, while the effort-response component attenuates more steeply, by γ^2, because the multiplier enters the effort margin twice: a weaker chain dampens how much each unit of effort raises coverage, and it also blunts the mayor's incentive to exert effort in the first place. Under the assumption that ψ_0 is small relative to $\bar{\psi}$, the ratio of inactive- to active-subgroup first-stage slopes is bounded between γ and γ^2. A within-subgroup first-stage ratio close to one rejects the augmented model; any substantially below one is consistent with it and provides a range estimate of γ.

The multiplier $g(\kappa^I_m, \kappa^A_m) = \gamma + (1-\gamma)\kappa^I_m \kappa^A_m$ treats the two capacity margins symmetrically: a single γ governs the attenuation when either component is inactive. The framework therefore makes no prediction that infrastructure and alignment must attenuate at the same rate; if the data return distinct γ_{κ^I} and γ_{κ^A}, this should be read as an empirical refinement consistent with the framework rather than as an additional model prediction.

The augmented model preserves the benchmark's headline-level predictions, both a positive first stage on the full sample and a positive coefficient on vote share, but predicts a sharply different shape of heterogeneity. Where the benchmark predicts smooth attenuation across subgroups, the augmented model predicts that the first stage attenuates to near its floor γ where any structural margin is missing. In municipalities without a pre-existing *CRAS*, the implementation chain has no apparatus to translate authorized slots into realized expansion. Inside *CRAS* municipalities, the same logic applies to manager turnover: where the front-line operator was not replaced after the

2008 municipal election, the chain operates at the floor productivity γ rather than at full intensity. The active cell, where both margins are simultaneously present, is where the headline coefficient is essentially identified.

3.4 Voter Response

We close the model by characterizing how realized expansion translates into vote share. Voters do not observe the mayor's type θ_m directly; instead, they form beliefs about competence from observable program implementation outcomes, in the spirit of the retrospective-voting tradition of Ferejohn (1986) and the more recent literature on voters learning from government performance in developing democracies (Ferraz and Finan, 2008; Avis et al., 2018). Each voter i in municipality m observes a noisy signal

$$s_{im} = \nu_0 + \nu_1 d_{im} + \nu_2 e_m + \varepsilon_{im}, \qquad \nu_1, \nu_2 > 0, \tag{12}$$

where ε_{im} is mean-zero noise independent across voters and conditional on e_m. The first informative component, $\nu_1 d_{im}$, captures the voter's direct experience as a beneficiary. The second, $\nu_2 e_m$, captures the broader visibility of local implementation as registration campaigns, outreach activity, and welfare-office operations make program delivery observable beyond direct beneficiaries.

Bayesian updating closes the model in the standard career-concerns and political-agency tradition of Holmström (1999) and Maskin and Tirole (2004), surveyed by Ashworth (2012). Voters apply Bayes' rule using the priors $p_0 = \Pr(\theta_m = 1)$ and the conditional signal densities f_0 and f_1:

$$\Pr(\theta_m = 1 \mid s_{im}) = \frac{f_1(s_{im})p_0}{f_1(s_{im})p_0 + f_0(s_{im})(1-p_0)}. \tag{13}$$

By Proposition 1, competent mayors choose strictly higher a_m^* than incompetent mayors and therefore generate higher e_m^*. This shifts the conditional mean of s_{im} upward under $\theta_m = 1$. We assume that the noise term has a log-concave density, a standard regularity condition satisfied by common distributions such as the normal and logistic. Under this condition, the location shift induced by higher implementation preserves the monotone likelihood ratio property:

$$\frac{f_1(s)}{f_0(s)} \quad \text{is strictly increasing in } s. \tag{14}$$

Under (14), the posterior in (13) is strictly increasing in s_{im}. Voter inference is rational:

a higher signal does indicate higher probability of competence, because in equilibrium higher signals are more likely to have been generated by competent mayors. Aggregating across voters in the probabilistic-voting tradition of Lindbeck and Weibull (1987), with η_{im} smoothing across the ideology distribution, the linear vote-share form posited in equation (3) is recovered as the equilibrium aggregate. The aggregation parameter $\lambda_1 > 0$ collects the material channel, and $\lambda_2 \geq 0$ collects the inference channel across all voters, beneficiaries and non-beneficiaries alike).

Remark 1 (The two channels and the sign of the headline)**.** *Equation* (3) *predicts* $\partial V_m/\partial Q_m > 0$ *whenever* $\partial B_m^*/\partial Q_m > 0$ (*Proposition* 1 *or* 2) *and either* $\lambda_1 > 0$ *or* $\lambda_2 > 0$*. The sign of the headline coefficient therefore tests jointly whether the first stage operates and whether voters reward implementation.*

Partisan attribution and the inference channel. The inference parameter λ_2 aggregates the rate at which voters convert observed implementation into a posterior on the mayor's competence. A natural extension lets λ_2 vary across municipalities with the alignment between the local executive's partisan brand and the federal program's institutional sponsor. When the mayor's party is also the federal incumbent that designed and operates the program at the national level, the credit chain runs cleanly from local registration to national policy under a single partisan label; voters' Bayesian update on competence is sharper because the inferred type controls both ends of the implementation chain. Formally, decompose the inference parameter as $\lambda_{2,m} = \lambda_2^{\text{base}} + \lambda_2^{\text{attrib}} \cdot \mathbf{1}\{\text{partisan alignment}_m\}$, with $\lambda_2^{\text{attrib}} \geq 0$. The augmented vote-share function becomes

$$V_m = V_0 + \lambda_1 B_m + \lambda_{2,m} e_m, \tag{15}$$

and the per-beneficiary electoral return is larger under aligned mayors by $\lambda_2^{\text{attrib}}$. The model thus predicts that, holding realized expansion fixed, a municipality governed by a mayor whose party owns the federal program returns a higher per-beneficiary vote share than one governed by a non-aligned mayor. This prediction is testable on the partition that separates Workers' Party mayors from the rest, under the identifying restriction that the material channel λ_1 is partisan-invariant; the restriction is institutional, since the federally set benefit value and eligibility rules make the marginal contribution of a beneficiary to $\alpha\, d_{im}$ independent of local partisan alignment. The partisan multiplier is a feature of the model rather than an unmodeled regularity.

4 Empirical Strategy

Our empirical strategy exploits a 2009 federal reform that revised the formula used to compute municipal *Bolsa Família* quotas. The reform generated plausibly exogenous variation in municipalities' authorized beneficiary ceilings, but turning that authorization into actual enrollment depended on local implementation. The wedge between what each municipality was allowed to do and what it actually did is what lets us identify the political return to expansion. We use the formula-driven component of the quota change, recentered against a counterfactual quota built from the pre-reform rule (Borusyak and Hull, 2023), as an exogenous shock instrumenting realized beneficiary expansion in a two-period municipal panel; the design identifies an average causal response of incumbent vote share to enrollment (Angrist and Imbens, 1995). Identification rests on the exogeneity of the recentered shock, the two periods difference out fixed municipal heterogeneity and common electoral shocks, while the causal content comes from the instrument.

4.1 The 2009 Quota Reform as Source of Identification

The methodology for computing municipal quotas was implemented by the *Ministério do Desenvolvimento Social* (MDS) based on census and survey data, following a standardized algorithm with no discretion at either the local or federal executive level. Brollo et al. (2020) provide direct evidence supporting this claim. They find evidence of manipulation in the enforcement of program conditionalities but no indication of political influence over the allocation of quotas themselves.[7]

To isolate the variation attributable solely to the change in allocation rules, we construct counterfactual 2009 quotas by applying the pre-reform methodology, which had been used to set the 2004 and 2006 quotas, to the data that would have been available immediately before the reform,

$$\text{CountQuota}_m^{2009} = \frac{\text{Poor}_m^{2000} \cdot n_m^{[2000,2006]}}{\sum_{k \in s} \text{Poor}_k^{2000} \cdot n_k^{[2000,2006]}} \cdot 1.18 \cdot \text{Poor}_s^{2006} \tag{16}$$

where Poor_m^{2000} is the number of poor families in municipality m of state s from the

[7]Gerard et al. (2025) note that they cannot exactly reproduce the official MDS quota numbers using publicly available information and the pre-reform allocation formula. They do not interpret this discrepancy as evidence of political manipulation, and a more plausible explanation is that the remaining differences reflect unpublished administrative adjustments or small differences in auxiliary estimates used to operationalize the formula.

2000 Census, $n_m^{[2000,2006]}$ is an IBGE estimate of municipal population growth between 2000 and 2006, Poor_s^{2006} is the state-level estimate of poor families from the 2006 *Pesquisa Nacional por Amostra de Domicílios* (PNAD), and the factor 1.18 reflects a uniform expansion applied to all municipal quotas as part of the reform.[8] The actual 2009 quotas were computed by MDS using a revised methodology that incorporated updated poverty estimates and new income-volatility adjustments, and the exact formula is not publicly available. The preferred instrument is therefore

$$\Delta\text{Quota}_m^{2009} = \text{Quota}_m^{2009} - \text{CountQuota}_m^{2009} \tag{17}$$

This measure captures the cross-municipal variation in authorized expansion arising from the change in the federal allocation methodology, net of the uniform national expansion and local poverty differences. In the preferred specification, the counterfactual quota counts as poor any household with monthly per-capita income below R$90 in the 2000 Census and below R$137 in the 2006 PNAD. The R$137 cutoff is the eligibility threshold announced by MDS *Informe* nº 169 as part of the 2009 reform itself. The instrument's recentering by the formula-based counterfactual follows the identification approach of Borusyak and Hull (2023), in which the residual variation in ΔQuota_m is purged of the deterministic exposure component that would otherwise transmit through unrelated channels.[9]

4.2 Research Design

The design compares changes in incumbent electoral performance between 2008 and 2012 across municipalities that faced different intensities of authorized quota expansion. Because treatment varies in intensity and no municipality is fully untreated, identification follows the average-causal-response framework of Angrist and Imbens (1995): the 2SLS coefficient recovers a weighted average of marginal electoral responses to instrument-induced changes in realized enrollment. Three assumptions carry the argument, and we discuss each in detail below.

The primary challenge is the endogeneity of realized enrollment. Mayors influ-

[8]The last officially reported quota update before the 2009 reform used information from the 2000 Census and the 2004 PNAD. To construct the counterfactual, we update the pre-reform formula using 2006 PNAD data, which is the most recent information that would have been available to policymakers at the time, while holding the allocation rule fixed.

[9]The construction of the counterfactual quota follows Gerard et al. (2025). Because the R$137 eligibility line was introduced jointly with the 2009 expansion, we also build alternative versions of the instrument using the pre-reform R$120 line, which isolates the change in allocation methodology from the contemporaneous threshold change.

ence how many families are actually registered, so coverage growth reflects local political incentives. Municipalities where incumbents face stronger electoral pressures may expand coverage more aggressively precisely because the expected political return is higher (Pereira et al., 2020). We address this by using the exogenous federal authorization shock as an instrument for realized beneficiary expansion. The data are organized as a balanced municipality-by-election-year panel covering the 2008 and 2012 elections, and we estimate the system

$$\text{Beneficiaries}_{mt} = \alpha_m + \lambda_t + \pi_1 \left(\Delta\text{Quota}_m \times \text{Post}_t\right) + X'_{mt}\pi_2 + \eta_{mt} \tag{18}$$

$$\text{VoteShare}_{mt} = \alpha_m + \lambda_t + \beta_1 \widehat{\text{Beneficiaries}}_{mt} + X'_{mt}\beta_2 + \varepsilon_{mt} \tag{19}$$

where α_m and λ_t are municipality and year fixed effects, Post_t is an indicator for the 2012 election, and X_{mt} bundles the incumbent-party fixed effects and the Census 2000 baseline characteristics interacted with Post_t. The variable $\text{Beneficiaries}_{mt}$ is the twelve-month moving average of enrolled families in municipality m measured at election year t. The instrument ΔQuota_m is the time-invariant 2009 quota shock, and it enters as $\Delta\text{Quota}_m \times \text{Post}_t$ because the shock itself is a single number per municipality and would otherwise be absorbed by the municipality fixed effects.

Equations (18) and (19) describe the data-generating process at the panel level, but the estimator we implement reads more transparently in first differences. With two periods, the municipality fixed effects α_m are time-invariant within municipality and drop out of the within-municipality demeaning, leaving the inter-election change as the operative variation. The system reduces to a cross-sectional IV at the municipality level,

$$\Delta\text{VoteShare}_m \;=\; \Delta\lambda + \beta_1\, \Delta\widehat{\text{Beneficiaries}}_m + \Delta X'_m\beta_2 + \Delta\varepsilon_m \tag{20}$$

where $\Delta\text{Beneficiaries}_m$ is the change in the twelve-month moving average of enrolled families between the 2008 and 2012 elections, and $\Delta\lambda$ captures the election-cycle shift common to all municipalities. The coefficient β_1 identifies the marginal effect of the municipality-level change in beneficiaries between elections on the corresponding change in vote share, with the quota shock ΔQuota_m instrumenting the endogenous portion of $\Delta\text{Beneficiaries}_m$.

The panel and first-difference displays are two equivalent ways to organize the same design, but the first-difference version makes the estimand transparent. The object of interest is the political return to beneficiary expansion between elections: how changes in realized enrollment from 2008 to 2012 translate into changes in incumbent vote share. For this reason, equation (20) is the specification we take to the data. An alternative ANCOVA-style specification would instead instrument the 2012 level of beneficiaries, absorbing the 2008 level as part of the baseline. That target is less natural in this setting because pre-reform coverage is positively correlated with the 2009 quota shock: municipalities with higher baseline coverage tended to lie on the same poverty dimensions that the reform reweighted. As a result, a post-level specification partially loads the treatment on pre-existing program scale rather than on the expansion induced by the reform. Appendix C.9 reports this alternative specification.

Both $\text{Beneficiaries}_{mt}$ and ΔQuota_m are standardized to mean zero and unit variance within the estimation sample, so coefficients in the tables read as the response to a one-standard-deviation move in the relevant variable. The two natural-unit scales that anchor the substantive interpretation are the municipality-level "additional families" quantities reported in Panel B of Table 1.

Equation (20) therefore compares two municipality-level changes: additional realized beneficiaries between elections, instrumented by the additional authorized slots generated by the 2009 reform. The tables report the estimates in standardized units. In the text, we translate them into natural units using the inter-election scale of realized expansion: one standard deviation in $\Delta\text{Beneficiaries}_m$ corresponds to 947 additional families, so per-thousand-family effects are obtained by rescaling the standardized coefficient accordingly.[10]

Treatment varies continuously in intensity rather than in binary status, so the design identifies an average causal response (ACR) parameter in the sense of Angrist and Imbens (1995). The ACR is a weighted average of the marginal effect of expansion on the change in vote share, where the non-negative weights $\omega(d)$ are proportional to $\text{Cov}(\mathbf{1}\{D_m \geq d\}, Z_m)$ and integrate to one. Under the assumptions of independence, exclusion, and monotonicity, β_1 is the population analog of $\int \omega(d)\, \mathbb{E}[\partial Y(d)/\partial d]\, dd$. Realized expansion is right-skewed, so these weights concentrate among municipalities where the 2009 reform translated into substantive enrollment changes, and the population weights further re-weight the estimand toward the experience of the typical Brazilian citizen rather than the typical municipality. Because no municipality is fully untreated in either period, the design does not identify level treatment effects (ATT or ATE), and β_1 is the marginal response, not the level response.[11]

[10]Standardization is an affine transformation and does not affect t-statistics, p-values, the Kleibergen-Paap first-stage F, or Anderson-Rubin confidence sets. The panel variable $\text{Beneficiaries}_{mt}$ is standardized using its panel-level dispersion, 4,949 families. Since the first-difference object is the inter-election change, whose cross-municipal standard deviation is 947 families, the conversion factor from the panel-standardized coefficient to the inter-election scale is $947/4{,}949 \approx 0.19$.

[11]The recent panel-DiD literature uses the same "ACR" label for related parameters identified under strong parallel-trends assumptions at every dose level (notably Callaway et al., 2024). We do not invoke

Outcome. The main outcome is the incumbent's vote share,

$$\text{VoteShare}_{mt} = \frac{\text{IncumbentVotes}_{mt}}{\text{VotesOnCandidates}_{mt}}$$

where $\text{VotesOnCandidates}_{mt}$ is the total number of valid votes cast for any candidate in municipality m in election year t, excluding blank and null ballots. This normalization ensures that the outcome reflects the incumbent's competitive position among voters who expressed a preference for a candidate.

Sample. The estimation sample is restricted to municipalities in which the incumbent mayor elected in 2008 sought reelection in 2012. After excluding municipalities where the incumbent did not seek reelection, those with missing treatment, outcome, or baseline covariate information, and those without contested elections in both periods, the final sample contains 2,105 of the 5,568 Brazilian municipalities (approximately 38 percent), observed in two election years and yielding 4,210 municipality-year observations. Because the outcome is defined at the incumbent level, this restriction is unavoidable. Section 4.3.1 shows that the instrument does not predict rerunning, and Table 2 confirms that the estimation sample is broadly representative of the full population of municipalities.

Specification. All specifications include municipality and year fixed effects. Because municipality fixed effects absorb time-invariant municipal characteristics, the baseline controls enter only as interactions with the post indicator. In our preferred specification we additionally include incumbent-party fixed effects based on the pre-reform winner's party, which absorb party-specific electoral dynamics, and a bundle of municipality-level baseline characteristics from the 2000 Census (log population, population growth, share of poor, household income per capita, and urban share), which absorb any differential 2012 effect operating through pre-reform municipal conditions. We do not include the incumbent's 2008 vote share from the preferred specification to avoid introducing a lagged dependent variable. Combining unit fixed effects with a lagged dependent variable in a short panel is known to generate bias (Nickell, 1981; Imai and Kim, 2019).

All regressions are weighted by municipal population in 2000. This is a scope condition rather than a precision choice. The estimand targets the causal response for the typical Brazilian citizen, which, given Brazil's urban concentration, weights toward larger

this approach. Angrist and Imbens's ACR is identified through instrument exogeneity and monotonicity rather than through dose-level parallel trends.

municipalities where the absolute scale of expansion was economically meaningful.[12]

Standard errors are clustered at the state level, following Abadie et al. (2023), because the counterfactual quota formula incorporates a state-level poverty estimate and induces within-state correlation in the instrument.[13]

4.3 Threats to Identification

4.3.1 Conditional Independence

Assumption 1 (Conditional Independence)**.** *Conditional on municipality fixed effects, year fixed effects, incumbent-party fixed effects, and the baseline Census 2000 characteristics indicator, the quota shock is orthogonal to potential electoral outcomes:*

$$\mathbb{E}\left[\, Y_{m,t}(0) \;\middle|\; \Delta\text{Quota}_m \times \text{Post}_t,\ \alpha_m,\ \lambda_t,\ \phi_m^{2008},\ X_m^{2000} \times \text{Post}_t \,\right] \\ = \mathbb{E}\left[\, Y_{m,t}(0) \;\middle|\; \alpha_m,\ \lambda_t,\ \phi_m^{2008},\ X_m^{2000} \times \text{Post}_t \,\right].$$

The quota shock must be orthogonal to potential electoral outcomes conditional on municipality fixed effects, year fixed effects, incumbent-party fixed effects, and the baseline characteristics. Conditional rather than unconditional exogeneity is the relevant requirement here because the counterfactual quota formula incorporates a state-level poverty estimate, Poor_s^{2006}, which mechanically correlates the instrument with pre-existing political geography. Municipalities in poorer states receive systematically different quota shocks, and those same states have structurally distinct electoral dynamics (Zucco, 2013). Once the controls absorb time-invariant municipal characteristics, party-specific electoral baselines, and any 2012 trend operating through pre-reform municipal conditions, the residual variation in ΔQuota_m reflects the change in the federal allocation formula net of these dimensions and is plausibly orthogonal to potential electoral outcome trends.

The proportional-selection exercise adapts Oster (2019) as a stability-style sensitivity check on the reduced-form coefficient of the quota shock. Oster's framework was

[12]The right-skewed distribution of realized *Bolsa Família* expansion would otherwise give dominant weight to municipalities where expansion was modest and administrative capacity limited. Population weights align the estimand with the municipalities where the mechanism is most likely to operate. Unweighted and alternative weights' estimates are reported in Appendix C.3.

[13]Brazil is divided into 26 states plus the Federal District. The Federal District does not hold municipal elections and falls outside our sample, so the regressions cluster on 26 state-level groups. Even at this count, asymptotic cluster-robust inference can be unreliable when the first stage is strong. We therefore report Anderson and Rubin (1949) confidence sets throughout, which are valid under arbitrary instrument weakness and robust to the cluster count. Appendix C.4 also reports wild cluster bootstrap p-values (Cameron et al., 2008; MacKinnon et al., 2023).

developed for OLS on a single treatment, so we treat the resulting δ^* as a coefficient-stability heuristic rather than a structural bound: under instrument exogeneity the reduced form is causal on its own, and the proportional-selection assumption is invoked here only as a discipline on whether the observed coefficient is robust to the kind of selection that the observables already capture. Adding the Census 2000 baseline characteristics moves the coefficient on the quota shock from 0.64 to 0.61. The decline is small and consistent with limited selection on observables. Under the conservative bound $\overline{R}^2 = 1.3 \cdot R^2_{\text{long}}$ we obtain $\delta^* = 24.12$, and under Oster's recommended benchmark $\overline{R}^2 = 2.2 \cdot R^2_{\text{long}}$ we obtain $\delta^* = 6.03$. Read as a heuristic, for proportional selection on unobservables to overturn the reduced-form effect under the standard bound, unobservables would have to be at least six times as important as the rich set of observables we condition on.

Selection into rerunning. The estimation sample is restricted to municipalities where the 2008 incumbent sought reelection in 2012, so the instrument could bias inference if it systematically affected the decision to run again. Table 2 regresses an indicator for whether the 2008 incumbent appears on the 2012 ballot on the quota shock, estimated on the full universe of municipalities. Under the preferred counterfactual-based instrument, the identifying variation appears orthogonal to selection into the estimation sample.

Table 2: Instrument and the Incumbent's Decision to Run Again in 2012

	Incumbent Run in 2012		
	(1)	(2)	(3)
ΔQuota_m $(\hat{\gamma}_1)$	-0.005 (0.003)	-0.004 (0.004)	-0.003 (0.003)
Party Fixed Effects		✓	✓
Baseline Characteristics			✓
Observations	5,504	5,504	5,504

Notes: The dependent variable is an indicator equal to one if the 2008 incumbent appears on the ballot in the 2012 municipal election. The regression is estimated on the universe of municipalities, using the preferred instrument. Column (1) is bivariate; column (2) adds party fixed effects based on the 2008 winner; column (3) further adds baseline characteristics (Census 2000 log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula). Column (3) is the preferred specification, matching the control set of the main DiD-IV. Robust standard errors clustered at the state level are reported in parentheses. All regressions are weighted by population in 2000. 64 observations were excluded due to missing values in one or more variables.

As a complementary check, Section 5.4 re-estimates the preferred specification weighted by the inverse probability of selection into the rerunning sample. The 2SLS point estimate moves from 4.99 to 4.46 and remains statistically significant, which indicates that selection into the rerunning sample is not materially distorting the estimates.

Pre-reform electoral balance. A second exercise tests whether the quota shock is correlated with pre-existing political conditions. Table 3 reports placebo regressions of 2008 electoral outcomes on the preferred instrument within the estimation sample. Columns 1 to 3 use the 2008-winner's vote share as the outcome, and columns 4 to 6 use an indicator for the sitting incumbent winning in 2008. Because 2008 outcomes were determined before the reform, any systematic non-zero association would signal baseline imbalance rather than the effect of quota-induced expansion.

Table 3: Placebo: 2008 Electoral Outcomes on Post-Reform Quota Shock

	Vote Share			Winning		
	(1)	(2)	(3)	(4)	(5)	(6)
ΔQuota_m ($\hat{\delta}_1$)	-0.110 (0.140)	-0.112 (0.150)	-0.126 (0.168)	-0.005 (0.005)	-0.020 (0.022)	0.005 (0.012)
Party Fixed Effects		✓	✓		✓	✓
Baseline Characteristics			✓			✓
Observations	2,105	2,105	2,105	2,105	2,105	2,105

Notes: The dependent variable is measured in the 2008 municipal election. Vote Share is the 2008-winner's share of valid votes. Winning is an indicator equal to 100 if the sitting incumbent (2004-winner) won in 2008 and 0 otherwise. All columns use the preferred instrument and restrict to the municipalities in the DiD-IV estimation sample. Columns (1) and (4) are bivariate; columns (2) and (5) add party fixed effects based on the 2008 winner; columns (3) and (6) further add baseline characteristics (Census 2000 log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula). Robust standard errors clustered at the state level are reported in parentheses. All regressions are weighted by population in 2000.

The evidence is consistent with conditional orthogonality of the instrument. Point estimates remain near zero and statistically insignificant as progressively richer control sets are added, and none of the six specifications rejects the null of zero association at conventional levels. The instrument does not display the kind of pre-reform imbalance that would threaten identification.

No anticipation. The 2009 quota revision was formally announced in MDS *Informe nº 169*, dated April 2009, which communicated that approximately 1,300,000 additional families would be incorporated in staggered waves over the course of the year. The previous quota adjustment had occurred in 2006. Although municipalities may have anticipated that some revision would eventually take place, they could not have foreseen the magnitude, the distributional formula, or the timing of the 2009 reallocation before the official announcement, which came six months after the October 2008 municipal elections. The placebo balance reported just above is consistent with this institutional timeline, since the instrument is statistically unrelated to either the 2008 winner's vote share or the sitting incumbent's 2008 win probability in any specification.

4.3.2 Exclusion Restriction

Assumption 2 (Exclusion Restriction)**.** *The quota shock affects incumbent vote share only through realized beneficiary enrollment:*

$$\Delta\text{Quota}_m \times \text{Post}_t \;\perp\!\!\!\perp\; Y_{m,t}(\text{Beneficiaries}_{m,t}) \\ \Big|\; \text{Beneficiaries}_{m,t},\; \alpha_m,\; \lambda_t,\; \phi_m^{2008},\; X_m^{2000} \times \text{Post}_t.$$

Equivalently, conditional on the controls and on realized expansion, the quota shock has no direct effect on the incumbent's vote share.

This is fundamentally a substantive rather than a statistical claim, and the case for it rests on the institutional channel through which the shock operates. The 2009 reallocation was a federally administered formula change, computed centrally by the *Ministério do Desenvolvimento Social* from Census 2000 and PNAD 2006 data, and it modified one and only one variable observable to municipalities and to voters, namely the authorized number of beneficiary slots. The benefit value paid per family was set by separate federal decree, and conditionality enforcement was implemented by federal partners through schools and health facilities.

The institutional argument is the primary defense of the exclusion restriction, but the most plausible alternative channels have empirical counterparts that we can test directly. Two channels are worth addressing because they could, in principle, move with the same demographic inputs that enter the quota reform: the average per-family *Bolsa Família* benefit value and federal revenue-sharing transfers to municipalities (*Fundo de Participação dos Municípios*). The benefit schedule itself is set nationally and is not chosen by mayors, but the average benefit paid in a municipality can vary with family composition, since households with more children or adolescents receive additional variable benefits. If the quota shock is correlated with local demographic composition, this could mechanically generate changes in average benefit values even holding enrollment fixed. *Fundo de Participação dos Municípios* (FPM) is also relevant because its transfers are assigned using population-based rules; if the demographic estimates used in the 2009 quota reform are correlated with the population brackets, the quota shock could proxy for concurrent fiscal transfers rather than only for beneficiary expansion. Institutionally, however, neither channel is part of the *Bolsa Família* quota formula: benefit values are fixed by federal decree, and FPM coefficients are set by Complementary Law 91/1989 from population brackets and apportionment factors.

Table 4 reports the empirical exclusion tests. Panel A regresses the inter-election change in FPM transfers (interacted with post) on the standardized quota shock (interacted with post), under three control specifications matching the headline. Panel B reports the analogous test using the per-family *Bolsa Família* benefit value. In both panels, the preferred specification returns a coefficient statistically indistinguishable from zero. The empirical evidence is consistent with the institutional argument: the quota shock does not differentially shift per-family benefit values or concurrent FPM transfers across municipalities, so neither channel absorbs non-enrollment variation that would contaminate the IV.

Table 4: Instrument Exogeneity: FPM Transfers and Benefit Value

	(1)	(2)	(3)
Panel A: Δ FPM$_m$ $\times$ Post (std)			
ΔQuota$_m$ $\times$ Post ($\hat{\gamma}$)	-0.235 (0.197)	-0.213 (0.116)	0.151 (0.057)
Panel B: Δ Value$_m$ $\times$ Post (std)			
ΔQuota$_m$ $\times$ Post ($\hat{\gamma}$)	-0.006 (0.005)	-0.012 (0.005)	0.007 (0.003)
Party Fixed Effects		✓	✓
Baseline Characteristics			✓
Observations	4,210	4,210	4,210

Notes: Each panel reports the coefficient on the standardized 2009 quota shock interacted with post in a regression where the dependent variable is the indicated muni-level constant interacted with post. Panel A uses the inter-election change in *Fundo de Participação dos Municípios* transfers ($\Delta\,\text{FPM}_m = \text{FPM}_m^{2009} - \text{FPM}_m^{2008}$, standardized). Panel B uses the twelve-month moving average of the per-family *Bolsa Família* benefit value (standardized). Both regressions are estimated on the IV sample of 4,210 municipality-year observations using the preferred R$90/137 instrument. Column (1) includes only year and municipality fixed effects; column (2) adds incumbent-party fixed effects based on the pre-reform winner; column (3) further adds Census 2000 baseline characteristics interacted with the post indicator (log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor). All regressions are weighted by population in 2000. Robust standard errors clustered at the state level are reported in parentheses.

Reverse causality of realized expansion. Although it is conceptually distinct from the exclusion restriction, reverse causality between political strength and realized enrollment is the reason the design needs an instrument in the first place: if incumbents who were electorally stronger before the reform were also more able or willing to expand enrollment afterward, direct OLS estimates of the effect of realized expansion on vote share would be confounded.

Table 5: Pre-Reform Electoral Strength and Subsequent Realized Expansion

	Vote Share			Winning		
	(1)	(2)	(3)	(4)	(5)	(6)
Realized BF Expansion$_m$ ($\hat{\kappa}_1$)	-0.001	-0.005	-0.011	-0.041	-0.077	-0.007
	(0.059)	(0.044)	(0.013)	(0.023)	(0.044)	(0.014)
Party Fixed Effects		✓	✓		✓	✓
Baseline Characteristics			✓			✓
Observations	2,105	2,105	2,105	2,105	2,105	2,105

Notes: The dependent variable is realized BF beneficiary expansion between 2008 and the 12-month average of 2009–2012, standardized. The explanatory variable is incumbent electoral strength in 2008, measured alternatively as the 2008-winner's vote share (columns 1–3) or an indicator for the sitting incumbent winning in 2008 (columns 4–6). All columns restrict to the municipalities in the DiD-IV estimation sample. Columns (1) and (4) are bivariate; columns (2) and (5) add party fixed effects based on the 2008 winner; columns (3) and (6) further add baseline characteristics (Census 2000 log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula). Robust standard errors clustered at the state level are reported in parentheses. All regressions are weighted by population in 2000.

The estimates are small in magnitude and statistically indistinguishable from zero, so we find no robust evidence that pre-reform electoral strength predicts subsequent *Bolsa Família* expansion. This placebo result is reassuring, but it does not eliminate the broader endogeneity concern. Mayors may still expand enrollment more aggressively where they expect larger electoral returns, a form of reverse causality documented in Pereira et al. (2020). The federal quota shock addresses this concern by isolating variation in authorized expansion that is generated by the national allocation rule rather than by municipal political incentives. This is why we use the quota shock as an instrument for realized enrollment rather than relying on OLS comparisons of coverage growth.

4.3.3 Monotonicity

Assumption 3 (Weak Monotonicity)**.** *For every municipality m, a positive quota shock cannot reduce realized expansion:* $\Delta\text{Beneficiaries}_m(z') \geq \Delta\text{Beneficiaries}_m(z)$ *whenever* $z' > z$*, where* $\Delta\text{Beneficiaries}_m(z)$ *is the potential inter-election change in beneficiaries that municipality m would realize under shock value z.*

As in any setting, this is an identifying restriction rather than a testable hypothesis, since we never observe both potential outcomes for the same unit. We follow standard practice in the continuous-treatment IV literature (Imbens and Angrist, 1994; Angrist and Imbens, 1995) by combining an institutional argument with empirical diagnostics consistent with weak monotonicity. The diagnostics that follow bound the impact of unit-level violations on the average causal response, which is the object the empirical strategy targets; the unit-level assumption is sufficient for the bound but the bound itself relies only on the average property in the spirit of de Chaisemartin (2017).

The institutional argument is direct. The quota shock changes the amount of authorized room a municipality has to add beneficiaries; it does not mandate reductions in existing caseloads. Even municipalities with negative quota adjustments were not required, or legally allowed, to remove current beneficiaries simply because their authorized ceiling fell. Beneficiaries could be cancelled only through the program's regular eligibility and irregularity checks. A lower quota therefore restricts future incorporation: newly registered eligible families are less likely to be added to the payroll where authorized room is scarce. By contrast, municipalities receiving positive quota shocks acquire additional slots that can be converted into realized enrollment if local officials identify and register eligible households. The monotonicity claim is therefore about the direction of the opportunity set: more authorized room can relax the enrollment constraint, while less authorized room does not mechanically force beneficiary removal. A municipality that received a positive shock acquired headroom to enroll additional families if it chose to, while one that received a smaller adjustment lost no pre-existing capacity. There is no mechanical channel through which a higher federal authorization depresses enrollment. Defier behavior is in principle possible (a mayor could choose to scale back the registration apparatus precisely when the federal ceiling expanded), but we view this as empirically rather than institutionally implausible.

Three empirical diagnostics support this institutional reading. First, 67 percent of municipalities exhibit sign agreement between the inter-election change in beneficiaries and the quota shock, and these municipalities account for 98 percent of the IV-numerator mass $|\Delta D_m \cdot Z_m|$, which is the object that determines the 2SLS coefficient; the remaining 33 percent of sign-discordant municipalities are concentrated near zero on both axes (small expansion changes paired with small shocks) and contribute only 1.7 percent of the identifying mass. Second, a binned first-stage exercise tests whether the implementation chain runs in the same direction across the full support of the instrument: we sort municipalities into quintiles of Z_m and compute the conditional mean of ΔD_m within each bin. The conditional mean rises monotonically across the bulk of the distribution where the identifying variation concentrates, with the highest quintile (mean $Z_m \approx$ 1,446 additional authorized slots) producing roughly six times the inter-election expansion of the next-highest quintile. Third, Section 5.5 reports the first-stage Kleibergen-Paap F-statistic across three pre-specified partitions (Northeast, the in-Frey saturation cell, and the partisan PT/non-PT cut) and finds it positive and well above the weak-instrument threshold in every subgroup that identifies the IV, with no subgroup exhibiting a wrong-signed coefficient. This is a stronger version of average monotonicity than the binned check, since the first stage runs in the same direction not only across quintiles of the instrument but across institutionally meaningful partitions of the sample.

Adapting the sensitivity bound of de Chaisemartin (2017) to this continuous-treatment setting, sign-discordant municipalities contribute negative weight to the pooled coefficient and carry 1.7 percent of the absolute IV-numerator mass $|\Delta D_m \cdot Z_m|$. We construct the worst-case bound by assuming the sign-discordant municipalities have causal responses of magnitude equal to and direction opposite to the sign-concordant ones, the most adversarial defier configuration that the data leave room for: any larger defier response would have raised the share of negative implicit weights detected in Appendix D.3 above the level we measure. Under this worst case, the sign-concordant ACR is bounded above the headline by at most 0.09 percentage points (1.8 percent of the headline value), with the lower bound at the headline itself. The estimator is therefore essentially insensitive to monotonicity violations at any plausible magnitude.

4.3.4 The 2012 Quota Revision

Municipal authorization margins were revised again in 2012, shortly before the election.[14] This revision was considerably smaller than the 2009 expansion but may still have affected 2012 electoral outcomes. We treat the 2009 revision as the primary source of identifying variation because it is the first reform after the 2008 election and has the full inter-election period to affect registration efforts and voter perceptions. The 2012

[14]Following the release of the 2010 Census, the federal government updated municipal poverty estimates used to guide program expansion in 2012. See the MDS *Informe* nº 324.

update is both smaller in magnitude and much closer to the election. Appendix C.3 reports supplementary specifications that extend the measurement window for the endogenous variable through 2012 and an updated version of the instrument using information from the 2010 Census.

5 Main Results

We proceed in four steps. First, we establish that the 2009 quota reform generated substantial variation in realized program expansion (first stage). Second, we show that this exogenous variation in authorized quotas translates into better incumbent electoral performance: the intention-to-treat (ITT) effect of the quota shock on the ballot box, which is algebraically the reduced form. Third, we combine these two moments to recover the causal effect of instrumented beneficiary expansion on incumbent vote share and reelection (two-stage least squares). Fourth, we subject these estimates to a battery of robustness and heterogeneity checks. Throughout, both the year-varying treatment ($\text{Beneficiaries}_{mt}$) and the time-invariant instrument (ΔQuota_{m}) are standardized to mean zero and unit standard deviation within the estimation sample, so coefficients read as the response to a one-standard-deviation move in the relevant variable.

5.1 First Stage: Authorization and Coverage

Table 6 reports the first-stage relationship between the quota shock and realized beneficiary expansion. In the preferred specification, which adds incumbent-party fixed effects and the baseline characteristics, a one-standard-deviation increase in the quota shock raises year-varying standardized beneficiary enrollment by 0.198 standard deviations. The first-stage coefficient is 0.296 in Column (1), 0.269 in Column (2), and 0.198 in Column (3); the attenuation as baseline controls enter reflects cross-municipal variation in the 90/137 shock that is partially explained by the Census 2000.

The first-stage magnitude is economically substantial: one standard deviation of the shock corresponds to approximately 843 additional authorized beneficiaries, and the implied response in realized enrollment is approximately 980 additional families. In natural units the first-stage slope is therefore about 1.16 realized families per additional authorized beneficiary in the recentered shock. This is a projection coefficient, not a take-up rate, and is not bounded by one: the recentered instrument is one component of, and is positively correlated with, the broader 2009 authorization change that municipalities actually drew down, so the realized response per unit of the *measured*

Table 6: First-Stage Estimates

	Realized Expansion in 2009		
	(1)	(2)	(3)
ΔQuota_m (π_1)	0.296 (0.092)	0.269 (0.050)	0.198 (0.023)
Party Fixed Effects		✓	✓
Baseline Characteristics			✓
Observations	4,210	4,210	4,210

Notes: Each coefficient is estimated on the municipality-year panel of 2008 and 2012 elections for the 2,105 municipalities where the 2008 incumbent ran for reelection in 2012. The endogenous variable is the year-varying twelve-month moving average of *Bolsa Família* beneficiaries; the instrument is the time-invariant 2009 quota shock interacted with the post-2009 indicator. Under two-period two-way fixed effects, the coefficient identifies the marginal effect of the muni-level inter-election change in beneficiaries on the corresponding change in the outcome (first-difference identification). All specifications include year and municipality fixed effects. "Party Fixed Effects" are based on the pre-reform winner's party. "Baseline Characteristics" bundles five Census 2000 controls, each interacted with the post-reform indicator: log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula. All regressions are weighted by municipal population in 2000. Robust standard errors clustered at the state level (26 clusters) are reported in parentheses.

instrument need not lie below one.

5.2 Reduced Form: Authorization and Electoral Returns

Table 7 reports the intention-to-treat (ITT) effect of the 2009 quota reform on incumbent electoral performance. It is the most transparent test of whether being authorized to expand the program translates into electoral gains, and is causal under instrument exogeneity alone, since it does not require the exclusion restriction that the quota shock affects votes only through the realized beneficiary expansion channel. It is the effect of the federal *authorization* rather than the effect of the additional families actually enrolled.

In the preferred specification, a one-standard-deviation increase in the quota shock raises the incumbent's vote share by 0.986 percentage points and the probability of winning by 1.133 percentage points. The ITT effect on vote share moves from 0.949 in Column (1) to 1.209 in Column (2) and 0.986 in Column (3); the small step from Column (2) to Column (3) is consistent with limited selection on observables as discussed earlier. For winning, Columns (4) and (5) yield raw coefficients of 1.819 and 2.567, at-

Table 7: Reduced-Form Estimates

	Vote Share			Winning		
	(1)	(2)	(3)	(4)	(5)	(6)
ΔQuota_m (ρ_1)	0.949 (0.198)	1.209 (0.269)	0.986 (0.238)	1.819 (0.703)	2.567 (0.476)	1.133 (0.385)
Party Fixed Effects		✓	✓		✓	✓
Baseline Characteristics			✓			✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: Each coefficient is estimated on the municipality-year panel of 2008 and 2012 elections for the 2,105 municipalities where the 2008 incumbent ran for reelection in 2012. The endogenous variable is the year-varying twelve-month moving average of *Bolsa Família* beneficiaries; the instrument is the time-invariant 2009 quota shock interacted with the post-2009 indicator. Under two-period two-way fixed effects, the coefficient identifies the marginal effect of the muni-level inter-election change in beneficiaries on the corresponding change in the outcome (first-difference identification). All specifications include year and municipality fixed effects. "Party Fixed Effects" are based on the pre-reform winner's party. "Baseline Characteristics" bundles five Census 2000 controls, each interacted with the post-reform indicator: log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula. All regressions are weighted by municipal population in 2000. Robust standard errors clustered at the state level (26 clusters) are reported in parentheses.

tenuating to 1.133 once the baseline characteristics enter, consistent with pre-existing electoral heterogeneity inflating the unconditional ITT before the preferred specification nets it out.

5.3 Second Stage: Coverage and Electoral Returns

Table 8 reports two-stage least squares estimates of the effect of instrumented beneficiary expansion on incumbent electoral performance. Under the first-difference reduction in equation (20), the coefficient identifies the marginal effect of the municipality-level inter-election change in beneficiaries on the corresponding change in vote share.

In the preferred specification, a one-standard-deviation increase in instrumented beneficiary expansion raises the incumbent's vote share by 4.99 percentage points and the probability of reelection by 5.73 percentage points. For vote share, the Anderson-Rubin 95 percent confidence set is $[2.35, 8.45]$; for winning, the set is $[1.80, 10.05]$. Both sets exclude zero under arbitrary instrument weakness.

The standardized coefficient is useful for inference, but its substantive meaning is easier to see in family counts. The treatment variable is standardized using the panel-

Table 8: Two-Stage Least Squares Estimates

	Vote Share			Winning		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	3.206 (0.820)	4.491 (0.995)	4.989 (1.433)	6.141 (1.170)	9.536 (1.150)	5.732 (1.955)
Kleibergen-Paap F	10.30	29.09	74.16	10.30	29.09	74.16
AR 95% CI	[2.00, 7.15]	[2.65, 7.10]	[2.35, 8.45]	[2.65, 8.90]	[7.30, 12.45]	[1.80, 10.05]
Party Fixed Effects		✓	✓		✓	✓
Baseline Characteristics			✓			✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: Each coefficient is estimated on the municipality-year panel of 2008 and 2012 elections for the 2,105 municipalities where the 2008 incumbent ran for reelection in 2012. The endogenous variable is the year-varying twelve-month moving average of *Bolsa Família* beneficiaries; the instrument is the time-invariant 2009 quota shock interacted with the post-2009 indicator. Under two-period two-way fixed effects, the coefficient identifies the marginal effect of the muni-level inter-election change in beneficiaries on the corresponding change in the outcome (first-difference identification). All specifications include year and municipality fixed effects. "Party Fixed Effects" are based on the pre-reform winner's party. "Baseline Characteristics" bundles five Census 2000 controls, each interacted with the post-reform indicator: log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula. All regressions are weighted by municipal population in 2000. Robust standard errors clustered at the state level (26 clusters) are reported in parentheses. The Kleibergen-Paap F-statistic is the squared t-ratio of the quota shock in the first-stage regression with cluster-robust standard errors. Anderson-Rubin 95% confidence sets are robust to arbitrary instrument weakness. An alternative ANCOVA-style specification, with the endogenous variable interacted with the post indicator, is reported in Appendix C.9.

level dispersion of beneficiaries, 4,949 families. The policy-relevant change, however, is the cross-municipal inter-election expansion, whose standard deviation is 947 families. Rescaling the 2SLS coefficient by this inter-election dispersion implies that a typical one-standard-deviation expansion between 2008 and 2012 raises the incumbent's vote share by $4.99 \times 947/4{,}949 \approx 0.95$ percentage points. Equivalently, each additional 1,000 beneficiary families enrolled between the two elections raises incumbent vote share by approximately 1.01 percentage points. These are two natural-unit translations of the same panel-standardized coefficient.

The 2SLS estimates are materially larger than their OLS counterparts: the OLS coefficient on the same standardized endogenous variable is 2.26 percentage points on vote share, compared with 4.99 percentage points in the 2SLS estimate. A Wu-Hausman test rejects exogeneity at the one percent level for both vote share and the binary winning indicator.[15]

[15] The IV-OLS gap on vote share, around 2.7 percentage points, admits two complementary readings, both consistent with the data. The first is negative political selection: mayors facing stronger electoral pressure expand coverage more aggressively, generating a downward bias in OLS that the federal au-

The preferred estimate of 4.99 percentage points is large relative to the stakes of a municipal race. The typical inter-election decline in incumbent support in our sample is 3.00 percentage points, in line with the incumbency disadvantage that Klašnja and Titiunik (2017) document for Brazilian municipal elections using a regression-discontinuity design on close races. The estimate is a marginal, complier-weighted response rather than a level effect for the average incumbent, so it does not translate one-for-one into a population swing; but for a complier municipality that realizes a one-standard-deviation expansion, the induced gain is of the same order as, and exceeds, that average decline. The effect is large enough to be decisive in the typical close race: 26 percent of races in the analysis sample were decided by fewer than 5 percentage points in 2012, and 11 percent by fewer than 2 percentage points.

5.4 Robustness

Table 9 presents the core sensitivity analysis across six specifications. Column (1) reproduces the preferred estimate ($\hat{\beta}_1 = 4.99$). Column (2) adds the 2008 incumbent's vote share to the baseline-characteristics bundle, addressing the lagged-dependent-variable concern discussed earlier; the point estimate is 4.47 and remains significant, confirming that the headline result is not an artifact of excluding the lagged outcome. Columns (3) to (6) perturb the treatment, outcome, or sample while holding the instrument fixed.

Column (3) redefines the endogenous variable as the change in the 12-month beneficiary count between the close of 2008 and the close of 2012, measuring the expansion as near to the ballot as the data allow; it yields $\hat{\beta}_1 = 2.99$. The attenuation reflects the 2012 quota revision, which used the 2010 Census to recalibrate municipal poverty estimates and reduced authorized quotas in municipalities whose 2010 poverty figures fell below the 2009 projections, partially reversing the 2009 over-allocation in those places. A window running through December 2012 therefore averages the 2009 expansion with this partial 2012 reversal. We prefer the headline window because it aligns the treatment with the period the federal government actually assigned: the 2009 reform instructed municipalities to fill their newly authorized quotas during 2009. Columns (4) and (5) winsorize the outcome and use the total cast-ballots denominator, respectively;

thorization shock removes. The second is classical measurement error in beneficiary counts, due for example to administrative miscounts or lagged updates in monthly enrollment records, which attenuates the OLS slope toward zero and is purged by the IV's projection on the exogenous instrument. The two channels are not mutually exclusive, and the IV-OLS gap on the winning indicator is consistent with either or both. Appendix C.4 reports the full diagnostic suite, including the Kleibergen-Paap statistic, Anderson-Rubin sets, wild cluster bootstrap, leave-one-state-out, and inverse-probability weights.

Table 9: Robustness Checks: Vote Share

	Main	Lag VS 2008	Ext. Window	Winsorized	Alt. Denominator	IPW
Panel A: First Stage						
ΔQuota_m (π_1)	0.198 (0.023)	0.197 (0.023)	0.330 (0.052)	0.198 (0.023)	0.198 (0.023)	0.207 (0.018)
Panel B: Reduced Form						
ΔQuota_m (ρ_1)	0.986 (0.238)	0.880 (0.198)	0.986 (0.238)	0.875 (0.215)	0.913 (0.223)	0.921 (0.235)
Panel C: Two-Stage Least Squares						
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	4.989 (1.433)	4.473 (1.265)	2.986 (0.775)	4.429 (1.251)	4.618 (1.321)	4.458 (1.326)
Kleibergen-Paap F	74.16	73.50	40.21	74.16	74.16	137.39
AR 95% CI	[2.35, 8.45]	[2.20, 7.60]	[1.55, 4.90]	[2.10, 7.40]	[2.20, 7.80]	[2.00, 7.50]
Party Fixed Effects	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: All columns report the preferred specification: party fixed effects + baseline characteristics $\times$ post, population-2000 weights. The outcome is incumbent vote share except where noted. Column (2) adds the 2008 incumbent vote share $\times$ post to the baseline-characteristics bundle, testing sensitivity to including a lagged dependent variable (Nickell, 1981; Imai and Kim, 2019); the main specification omits this term. Column (3) uses the 12-month moving average of realized beneficiaries through December 2012 as the endogenous variable. Column (4) winsorizes the outcome at the 1st and 99th percentiles. Column (5) uses total votes cast (including blank and null) as the denominator. Column (6) weights by the inverse probability of selection into the rerunning sample. Anderson-Rubin 95% confidence sets are robust to arbitrary instrument weakness. Robust standard errors clustered at the state level in parentheses.

Column (6) reweights by the inverse probability of selection into the rerunning sample.

5.5 Heterogeneity

We close the main results with a heterogeneity exercise along three sample partitions: Northeast region, the population-by-HDI cell that anchors Frey (2019), and the 2008-elected incumbent's party. In each partition, the instrument and the endogenous variable are interacted with a binary indicator and its complement, yielding two coefficients per equation from a single full-sample regression. The decomposition tests the augmented model's central prediction that the first stage attenuates on partitions aligned with the structural margins the model identifies, and remains comparable on partitions orthogonal to them.

The Northeast partition (columns 1–3) leaves the first stage intact in both subsamples: the instrument is strong inside and outside the region. The electoral response, by contrast, concentrates outside the Northeast, where the 2SLS estimate tracks the pooled headline; inside the region it is statistically indistinguishable from zero. The headline is therefore identified primarily from non-Northeastern compliers. Region is not one

Table 10: Heterogeneity by Subgroup: Vote Share

	Northeast			Frey 2019			PT incumbent		
	NE	non-NE	Δ	in Frey	out Frey	Δ	PT	non-PT	Δ
Panel A: First Stage									
ΔQuota_m (π_1)	0.027 (0.062)	0.244 (0.031)	-0.217 (0.070)	0.009 (0.006)	0.197 (0.023)	-0.188 (0.024)	0.126 (0.035)	0.230 (0.027)	-0.104 (0.045)
Panel B: Reduced Form									
ΔQuota_m (ρ_1)	0.736 (0.771)	1.037 (0.273)	-0.301 (0.853)	-0.823 (1.185)	0.988 (0.237)	-1.812 (1.128)	0.872 (0.156)	1.023 (0.285)	-0.151 (0.286)
Panel C: Two-Stage Least Squares									
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	20.255 (54.851)	5.753 (3.765)	14.503 (51.296)	-244.924 (185.679)	5.371 (1.513)	-250.295 (185.847)	8.057 (3.120)	4.385 (1.561)	3.672 (2.138)
Kleibergen-Paap F	0.19	61.70		2.81	73.70		12.61	71.81	
AR 95% CI	[-15.00, 25.00]	[1.80, 7.65]		[-15.00, 25.00]	[2.40, 8.50]		[3.20, 19.65]	[1.75, 8.15]	
Party Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210	4,210	4,210	4,210

Notes: All estimates are obtained from a single full-sample dual-IV regression per heterogeneity dimension: the instrument and the endogenous variable are each interacted with a binary subgroup indicator and its complement, producing two coefficients per equation from one regression. Columns (1)–(3) split on Northeast region (AL, BA, CE, MA, PB, PE, PI, RN, SE) versus other regions. Columns (4)–(6) split on the Frey (2019) sample restriction (population in 2000 below 30,000 *and* Atlas IDHM 2010 below 0.7), with column (4) inside Frey's sample and column (5) outside. Columns (7)–(9) split on whether the 2008-elected mayor's party (`winner_party_2008`) — the sitting incumbent during the 2009–2012 expansion window — is the Workers' Party (PT) versus any other party. Δ is the Wald test of equality of the two coefficients reported as the difference with its standard error. Specification matches the headline tables: 90/137 quota shock instrument, year-varying realized beneficiaries (12-month MA, standardized) as the endogenous variable, party fixed effects (pre-reform winner) plus baseline characteristics $\times$ post (Census 2000 log population, share of poor, household income per capita, urban share, and 2000–2006 population growth factor), municipality and year fixed effects, weighted by 2000 population, with cluster-robust standard errors at the state level. Anderson-Rubin 95% confidence sets are computed by grid inversion holding the other group's coefficient at its dual-IV estimate; they are robust to weakness of the corresponding sub-instrument.

of the structural margins the augmented model predicts should govern identification.

The Frey partition (columns 4–6) is sharper and more telling. Inside the cell that anchors Frey (2019)'s identification the first stage collapses well below the conventional weak-instrument threshold; the corresponding 2SLS estimate is uninterpretable noise, and identification comes entirely from outside the cell, where the response again matches the pooled headline. The collapse is mechanical, not a failure of the instrument: only 4.1 percent of paper-sample municipalities inside the Frey cell are classified as slack on the Q1 saturation cut, against 55.7 percent outside it, so with utilization rates B_m/Quota_m already this high a positive authorization shock has no extensive margin to push into.

We are careful, however, about why that margin is missing. The saturation measures are built from 2008 data, before our 2009 shock but after the earlier, federally driven expansion that Frey (2019) studies; a high 2008 utilization rate in the Frey cell may therefore be a product of that prior expansion rather than a fixed structural ceiling. We read the contrast accordingly: not that these municipalities are intrinsically unable to respond, but that the federally controlled enrollment channel had already absorbed much of the eligible pool here, leaving no room for the 2009 authorization to operate. Because the partition is dated 2008, it is predetermined with respect to our shock and does not threaten identification; it simply locates our variation in a different region of the support. This sharpens the distinction between the two designs: Frey (2019) identifies the electoral return to a federally controlled enrollment channel in the regime where it still had room to grow, whereas we identify the return to a locally controlled channel where room remains.

The PT partition (columns 7–9) identifies a positive electoral return on both sides of the partisan divide. The Anderson-Rubin set excludes zero whether or not the 2008 incumbent is from the Workers' Party. The 2SLS estimates differ in level, 8.06 percentage points for PT against 4.39 for non-PT, but the difference is not statistically distinguishable.

The point estimates line up with the direction the framework anticipates. The asymmetry, such as it is, sits in the per-beneficiary return rather than in the first stage; if anything the conversion of authorization into realized enrollment is somewhat slower under PT mayors, yet the point estimate of the per-beneficiary return is larger. This is the shape the partisan-attribution extension of the voter-response block commits to in advance, raising the per-beneficiary return by $\lambda_2^{\text{attrib}} \geq 0$ wherever the mayor's party also operates the federal program while holding the first stage partisan-invariant because the material channel λ_1 is federally set.

The mechanism behind $\lambda_2^{\text{attrib}}$ is institutional. *Bolsa Família* was created in 2003 under the federal Lula administration and has been the Workers' Party's flagship anti-poverty policy ever since; the 2009 reform that supplies our identifying variation was implemented under a PT-controlled federal executive in Lula's second term, and the 2012 election that we study was held under the first PT presidency of Dilma Rousseff. A municipality with a PT mayor between 2008 and 2012 is therefore administering a program created and politically owned by the same party at the federal level, throughout a period of continuous PT control of the federal executive. Under these conditions voter attribution to the local incumbent should be cleanest: the partisan chain of credit runs from the local registration apparatus to the federal program design, while the same expansion under a non-PT mayor splits credit between local administration and federal sponsorship. We read the larger PT point estimate as suggestive of this attribution channel rather than as a separate structural margin, while noting that the difference is too imprecisely estimated to bear independent weight. The paper's central message does not depend on it: centralized program design does not, on its own, depoliticize redistribution, and the return we document holds regardless of the incumbent's party.

6 Possible Mechanisms

The main results establish that federally induced quota expansion generated electoral returns for incumbent mayors. This section uses the same framework to test the augmented model's structural predictions about which margins of capacity must be in place for the federal-to-local chain to operate. We organize the evidence around the two margins: infrastructure (κ^I) and alignment (κ^A). Throughout, the object is not the standalone causal effect of capacity itself, but the scope condition under which the federal authorization shock creates the realized-enrollment variation from which electoral returns can be identified.

6.1 Infrastructure: κ^I

Of the 2,105 municipalities in the estimation sample, 1,415 (67.2 percent) had at least one *CRAS* in the 2008 *Censo SUAS*; the remaining 690 did not. The augmented model is unambiguous about the implication: the within-subgroup first stage should attenuate by roughly a factor of γ in $\{\kappa^I_m = 0\}$, where γ is the productivity floor of the implementation chain. The benchmark, by contrast, predicts smooth attenuation of comparable magnitude. The decomposition that follows tests the two against each other.

The first stage and reduced form deliver the asymmetry the augmented model predicts. The CRAS subsample reproduces the pooled-sample first stage almost exactly, while the no-CRAS subsample retains a statistically strong but much smaller pass-through, roughly a fifth as large (0.039 against 0.197). The CRAS-subgroup reduced-form responses on vote share and reelection are essentially indistinguishable from the headline. In the no-CRAS cell the first stage is too weak relative to the residual variation to identify a 2SLS effect: the Anderson-Rubin set is unbounded, so the cell carries no interpretable causal content even though its reduced form on vote share is individually significant. The pattern is consistent with the broader finding in the *Bolsa Família* literature that implementation, not design, is where municipal governments shape distributive outcomes (Fried, 2012; Sugiyama and Hunter, 2013).

Table 11: CRAS Presence as Necessary Condition

	Vote Share			Winning		
	has = 1	has = 0	Δ	has = 1	has = 0	Δ
First Stage (π_1)	0.197	0.039	0.158	0.197	0.039	0.158
	(0.023)	(0.003)	(0.023)	(0.023)	(0.003)	(0.023)
Reduced Form (ρ_1)	1.000	-4.851	5.851	1.125	4.434	-3.309
	(0.238)	(1.750)	(1.623)	(0.386)	(7.305)	(7.335)
Two Stage Least Squares (β_1)	5.086			5.699		
	(1.443)			(1.972)		
Kleibergen-Paap F	74.55	195.30		74.55	195.30	
AR 95% CI	[2.45, 8.55]	$[-\infty, +\infty]$		[1.75, 10.05]	$[-\infty, +\infty]$	
Party Fixed Effects	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: All estimates are obtained from the full-sample dual-IV regression: the instrument and the endogenous variable are each interacted with $\mathbf{1}(\text{CRAS}_m^{2008})$ and its complement, producing two coefficients per equation from a single regression. The "has = 1" column reports the coefficient on the CRAS-interaction; "has = 0" reports the coefficient on the no-CRAS interaction. Δ is the Wald test of $\beta_{has=1} = \beta_{has=0}$ reported as the difference with its standard error and $\chi^2(1)$ p-value. Under the preferred 90/137 instrument the first stage is strong in *CRAS* municipalities ("has = 1") but collapses in magnitude where no *CRAS* existed in 2008 ("has = 0"): the no-*CRAS* 2SLS coefficient and its Δ Wald difference are not identified and are left blank, and the corresponding Anderson-Rubin 95% confidence set is unbounded. All specifications include year and municipality fixed effects, party fixed effects based on the pre-reform winner (2004-winner in 2008; 2008-winner in 2012), and baseline characteristics $\times$ post: log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor (all from Census 2000). Weights: municipal population in 2000. Cluster-robust standard errors at the state level in parentheses.

The 2SLS coefficients sharpen the asymmetry rather than add to it. Because *CRAS* presence is measured before the 2009 reform, the subgroup split preserves the same IV logic used in the pooled design: within each cell, the quota shock supplies exogenous variation in authorized expansion, and the first stage determines whether that authorization generates realized enrollment to learn from. The *CRAS*-subgroup point estimates and Anderson-Rubin confidence sets reproduce the pooled headline directly. The no-*CRAS* subgroup, by contrast, has too little first-stage pass-through to support a meaningful second-stage estimate; its AR set spans the admissible range, so the design is effectively silent about the per-beneficiary return there. The implied weights make this scope condition explicit. In the saturated specification, $\omega_{\text{CRAS}} \approx 0.93$ and $\omega_{\text{no-CRAS}} \approx 0.07$, so the identifying covariance behind the headline 2SLS is overwhelmingly concentrated in municipalities with a pre-existing *CRAS*.[16]

[16]The implicit-weight characterization of the pooled 2SLS as $\hat{\beta}_{\text{2SLS}} = \sum_g \omega_g \beta_g^{\text{FWL}}$ follows the frame-

The substantive content of the table is the scope condition implied by the augmented model. Federal authorization becomes realized enrollment only when the implementation apparatus is active; the estimates show that this margin is not merely a source of heterogeneity in effect size, but a condition for the causal chain to operate. Because *CRAS* presence is measured before the reform, the subgroup split asks where the same federally induced quota shock can be converted into realized enrollment. In municipalities without a pre-existing *CRAS*, the shock produces too little pass-through to support an interpretable per-beneficiary electoral return; the design is therefore silent about the return to a beneficiary in that cell. What it does identify is the prior implementation constraint: federal authorization generates the complier variation behind the headline estimate only where a local delivery apparatus exists. In this sense, *CRAS* presence is a necessary condition for this centralized expansion to produce electoral returns, because it is necessary for authorized slots to become actual beneficiaries at scale.

6.2 Alignment: κ^A

Conditional on the existence of this delivery apparatus, the next question is whether the apparatus is politically usable by the incoming mayor. The augmented model predicts that the alignment margin κ^A is the second binding constraint: the within-subgroup first stage should attenuate sharply in *CRAS* municipalities where the incoming mayor did not replace the front-line manager on taking office. Table 12 reports the corresponding decomposition: no *CRAS* in 2008, *CRAS* where the manager stayed, and *CRAS* where the new mayor replaced the manager. We define the manager as the *de facto* frontline operator: the respondent of the unit-level *Censo SUAS* questionnaire, that is the person who actually fills the form and runs the unit day-to-day. The 2008 and 2009 *Censo SUAS* are annual snapshots recorded just before and just after the mayor elected in 2008 takes office, so a change in the recorded manager between the two waves identifies a replacement made by the incoming administration; the *de facto* definition admits an exact CPF match across waves, and lets us classify changes as external hires versus internal transfers.

The first stage in *CRAS* municipalities with manager turnover reproduces the pooled headline almost exactly, while the other two cells collapse below the conven-

work of Goldsmith-Pinkham et al. (2022) on contamination in linear regressions with multiple treatment effects, and the closely related decomposition of Słoczyński (2022) for treatment-effect heterogeneity. Both papers show that when an estimator pools heterogeneous within-cell responses through residualized regressors, the recovered coefficient is a weighted average whose weights depend on the residualized covariance structure rather than on cell sizes.

Table 12: Heterogeneity by Manager Turnover

	Vote Share			Winning		
	non-CRAS	CRAS / non-Turn	CRAS / Turnover	non-CRAS	CRAS / non-Turn	CRAS / Turnover
First Stage (π_1)	0.039 (0.003)	-0.015 (0.032)	0.207 (0.023)	0.039 (0.003)	-0.015 (0.032)	0.207 (0.023)
Reduced Form (ρ_1)	-4.953 (1.656)	0.638 (1.137)	1.051 (0.249)	4.010 (7.302)	-1.673 (4.639)	1.493 (0.477)
Two Stage Least Squares (β_1)			4.921 (2.252)			7.942 (2.525)
Kleibergen-Paap F	193.34	0.20	80.99	193.34	0.20	80.99
AR 95% CI	$[-\infty, +\infty]$	$[-\infty, +\infty]$	$[+2.30, +8.40]$	$[-\infty, +\infty]$	$[-\infty, +\infty]$	$[+2.95, +13.20]$
Party Fixed Effects	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: All estimates are obtained from a single full-sample triple-IV regression per outcome. The instrument and the endogenous variable are each interacted with three mutually exclusive indicators, ordered in the table from left to right: non-CRAS (municipalities with no *CRAS* in the 2008 *Censo SUAS*); CRAS / non-Turn (municipalities with a *CRAS* in 2008 but no manager change between the 2008 and 2009 waves); and CRAS / Turnover (municipalities with a *CRAS* in 2008 and a manager change between the 2008 and 2009 waves). The manager is defined as the *de facto* frontline operator: the `cpf_agente` respondent of the unit-level *Censo SUAS* questionnaire, the person who actually fills the form and operates the unit day-to-day, who is not necessarily the formal *Coordenador*. All specifications include year and municipality fixed effects, party fixed effects based on the pre-reform winner (2004-winner in 2008; 2008-winner in 2012), and baseline characteristics $\times$ post: log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor (all from Census 2000). Weights: municipal population in 2000. Cluster-robust standard errors at the state level in parentheses. The 2SLS coefficient and Anderson-Rubin set are reported only for the CRAS / Turnover cell, the one subgroup whose first stage identifies the coefficient; in the non-CRAS and CRAS / non-Turn cells the first stage is too small in magnitude (or, for non-turnover, statistically weak) to pin down the level coefficient, so the 2SLS estimate is left blank and the Anderson-Rubin 95% set is unbounded, reported as $[-\infty, +\infty]$. Anderson-Rubin sets for the identified cells are obtained by grid inversion and are valid under arbitrary instrument weakness.

tional weak-instrument threshold.[17] The Anderson-Rubin confidence sets reported within each cell confirm the asymmetry directly: in the *CRAS*-with-turnover cell, the AR 95% set on vote share is [2.30, 8.40], comfortably bracketing the headline 4.99; in the *CRAS*-without-turnover cell, the AR set is pinned to the grid boundary, the appropriate inference output when the first stage is too weak to identify the within-cell return. The channel operates where the person who actually runs the unit changes.

The hire-type composition of those changes is informative. Across 3,357 manager changes in 2,583 municipalities, 75.5 percent involve a successor absent from the 2008

[17]Two Frisch-Waugh-Lovell decompositions appear in this section and they measure related objects. Splitting the full sample into *CRAS* versus no-*CRAS* places about 0.93 of the implicit identifying weight on the *CRAS* municipalities; refining the *CRAS* group by manager turnover, about 0.88 of the full-sample weight falls on the *CRAS*-with-turnover cell, with the *CRAS*-without-turnover cell carrying a small positive weight (about 0.05) and no-*CRAS* the remainder. The variance-share and implicit-coefficient versions of these weights nearly coincide, so no cell carries a negative implicit weight. The three-cell partition is defined on the 1,292 *CRAS* municipalities whose *de facto* manager can be matched by CPF across the 2008 and 2009 *Censo SUAS* waves; 123 *CRAS* municipalities without a matchable manager are not classifiable by turnover and are omitted from this refinement.

CRAS human-resources roster anywhere in the same municipality, 14.7 percent an internal transfer from another role at the same *CRAS*, and 9.8 percent a transfer from a different *CRAS* in the same municipality. The dominant case is fresh personnel, not internal reshuffles, which rules out the reading that the turnover indicator captures relabeling rather than genuine personnel inflows.

6.2.1 The Politics of Alignment

The decomposition locates the active implementation margin: within *CRAS* municipalities, the quota shock translates into realized expansion where the incoming administration replaced the inherited front-line manager. The next question is what this replacement margin represents. Are substitutes partisan loyalists selected into the post, bureaucrats who become politically aligned after appointment, or a different type of local actor whose subsequent behavior makes the office politically consequential?

Table 13 reports four reduced-form panels at the municipality level, using the *CRAS* and turnover definitions of Table 12. Each coefficient is the effect of a one-standard-deviation quota shock on the row outcome: management turnover and new *CRAS* openings in Panel A, and substitute-manager traits or actions in Panels B–D.

Does the shock induce management or infrastructure changes? Panel A reports a null on both margins: a one-standard-deviation shock does not significantly change the probability that a municipality replaces its *CRAS* manager, nor does it raise the probability that the municipality opens a new *CRAS*. Turnover is set by the 2008 mayoral transition, not by the federal shock; infrastructure expansion is not the mechanism either. The electoral return concentrates where the election-driven turnover occurred.

Was the substitute already politically tied to the mayor? Panel B says no. The shock does not significantly increase the probability that a turnover municipality receives a substitute previously affiliated with the mayor's party or coalition. The any-party row is positive, but that coefficient reflects a broader pool of politically affiliated substitutes rather than prior attachment to the incoming mayor.

What does the substitute do next? Panel C shows the main behavioral response: municipalities exposed to a larger shock are more likely to have a substitute who later runs for office. The coefficient is positive and marginally significant, and the mean is large enough to be substantively meaningful: about 11 percent of turnover municipalities have a substitute who contests an election between 2010 and 2016.

Does the substitute realign with the mayor? Panel D restricts attention to substitutes

Table 13: Manager Trajectory

	Municipality		
	β	Mean	N
Panel A. Does the shock induce management or infrastructure changes?			
Manager turnover	0.271 (0.285)	84.288	1,292
Opened a new *CRAS*	-2.097 (1.756)	8.551	1,415
Panel B. Does the shock select substitutes by prior affiliation?			
Affiliated with mayor's party	1.533 (1.023)	4.316	1,089
Affiliated with mayor's coalition	0.925 (0.895)	7.163	1,089
Affiliated with any party	3.168 (0.745)	17.539	1,089
Panel C. Does the shock raise the rate of running for office?			
Ran for office	1.928 (0.969)	10.927	1,089
Panel D. Does the shock align substitutes with the mayor?			
Joined mayor's party (2009–2012)	0.145 (0.133)	1.628	1,044
Joined mayor's coalition (2009–2012)	0.187 (0.137)	2.652	1,018

Each cell is a separate reduced-form regression of the row outcome on the standardized 2009 quota shock plus Census-2000 controls (log population, share in extreme poverty, household income per capita, urban share, and 2000–2006 population growth), with state fixed effects and population weights; standard errors (in parentheses) clustered at the state level. Coefficients in percentage points ($\times$100). The Mean column reports the sample mean of the row outcome, also in percentage points. All columns are at the municipality level, using the Table 12 definitions of *CRAS* and turnover. Panel A is estimated over municipalities with a pre-existing *CRAS*: the first row indicates any front-line manager replacement among incumbent units, and the second row indicates whether the municipality opened at least one new *CRAS* unit in 2009. Panels B–C are estimated over municipalities with manager turnover and indicate whether at least one substitute had the row trait. Panel D is estimated over municipalities with manager turnover and at least one substitute not previously affiliated with the relevant mayoral party or coalition; outcomes indicate whether at least one such at-risk substitute newly joined the mayor's party or coalition between 2009 and 2012.

not previously affiliated with the relevant mayoral party or coalition and asks whether they newly join it during the mayor's term. The estimates are small and statistically indistinguishable from zero for both the mayor's party and the mayor's coalition.

Table 13 therefore narrows the channel: the shock does not cause manager turnover or new infrastructure, does not select substitutes through prior mayoral ties, and does not induce partisan realignment after appointment. The response it does reveal, increasing with the size of the shock, is the subsequent electoral entry of the substitute.

Table 14: Substitute Manager Characteristics

Panel A. Comparison across managerial roles (incumbent units, 2008–2009)				
	# CPFs	% CPFs	Filiated	Ran
Stayer (same CPF 2008–2009)	270	91.5%	18.3%	5.8%
Outgoing (replaced in 2009)	1,224	89.7%	19.6%	6.5%
Incoming New (external hire)	951	87.4%	18.1%	10.8%
Incoming Old (internal transfer)	264	95.3%	9.7%	4.7%
Panel B. Offices sought by incoming managers who ran (TSE 2010–2016)				
	Level	# CPFs	Runners	All substitutes
City council (*vereador*)	Municipal	99	81.8%	8.14%
Vice mayor	Municipal	18	14.9%	1.48%
Mayor (*prefeito*)	Municipal	10	8.3%	0.82%
State deputy	State	2	1.7%	0.16%
Any office		121	100.0%	9.95%

Panel A: Each row aggregates manager-CPFs at incumbent *CRAS* units observed in both the 2008 and 2009 *Censo SUAS* waves. *Stayer* is the same CPF in both years; *Outgoing* is the 2008 manager replaced in 2009; *Incoming* is the 2009 substitute. # CPFs counts distinct managers within the role; % CPFs reports the share of within-role observations that map to a distinct manager (i.e., one minus the duplicate rate from managers serving multiple units in the same municipality). *Filiated* is the share of CPFs with an open party affiliation as of 1 January 2009. *Ran* is the share of CPFs that appear as a candidate for any elected post in the TSE 2010–2016 candidate database. *Panel B:* For substitutes who ran for office, the breakdown by office category. *Runners* reports each office's share of the 121 substitute managers who ran for any office; *All substitutes* reports each office's share of all substitute managers. CPFs running for multiple offices across 2010–2016 are counted once per office category, so office shares need not sum to 100 percent.

Table 14 then describes composition across manager types. Incoming external substitutes are no more likely to be party-affiliated than stayers or outgoing managers. Where they differ is subsequent electoral behavior. External incoming managers run for office at 10.8 percent, compared with 5.8 percent for stayers and 6.5 percent for outgoing managers. The candidacies are overwhelmingly local: of the 121 substitutes who ran, 99 stood for *vereador*, with smaller numbers for vice mayor (18), mayor (10), and state deputy (2); the office counts exceed 121 because some ran for more than one office across 2010–2016.

Taken together, the evidence points to a form of bureaucratic politicization closer to the career-platform mechanism in Colonnelli et al. (2020) than to loyalist allocation. We do not find that mayors use the expanded program to install prior co-partisans or coalition affiliates, nor that substitutes subsequently realign with the mayor's political bloc. What changes is the political trajectory attached to the post: external incoming managers are no more generally party-affiliated than stayers or outgoing managers, but they run for local office at roughly twice the rate of stayers, and this response increases with the size of the quota shock. This pattern closes the model's alignment channel. Once a functioning *CRAS* apparatus exists, replacing the inherited front-line manager can make that apparatus politically usable without requiring partisan selection at hiring; the multiplier on discretionary effort a_m rises, and federal authorization translates more strongly into realized expansion. The stepping-stone label therefore refers to the downstream political use of the managerial post, not to a claim about prior loyalty or the mayor's hiring motive.

7 Conclusion

This paper studies whether locally elected officials can obtain electoral returns from a welfare program whose formal design is centralized. *Bolsa Família* is an especially useful setting for this question because eligibility rules, benefit values, and payment flows are set nationally, while the registration process that turns eligible households into actual beneficiaries is administered locally. The 2009 quota reform generated municipality-level variation in authorized expansion, allowing us to compare places that received different amounts of federally created room to grow while holding constant the national program rules.

The main result is that authorized expansion translated into both realized enrollment and incumbent electoral gains. A one-standard-deviation increase in the quota shock raised incumbent vote share by about one percentage point in reduced form. Instrumenting realized beneficiary expansion with the quota shock, an additional 1,000 beneficiary families between the 2008 and 2012 municipal elections raised the incumbent mayor's vote share by approximately one percentage point. These estimates should be read as marginal, complier-weighted returns to the expansion induced by the reform.

The mechanism evidence shows that this return was conditional on local implementation capacity. The first margin is the existence of a functioning local delivery apparatus. Municipalities with a pre-existing *CRAS* converted the quota shock into realized enrollment at roughly the pooled first-stage rate, and their electoral response reproduces the headline estimates. Municipalities without a *CRAS* exhibit much weaker pass-through. The design is therefore not informative about the per-beneficiary electoral return in that subgroup; it is informative about the prior constraint. Federal authorization generated the enrollment variation behind the headline effect only where local administrative infrastructure was already in place.

The second margin is who operated that apparatus. Within *CRAS* municipalities, the identifying variation concentrates where the mayor elected in 2008 replaced the inherited front-line manager before the reform. The shock itself did not cause manager turnover or new *CRAS* openings. Nor does the evidence support a simple loyalist-allocation interpretation: substitute managers were not selected through prior ties to the mayor's party or coalition, and they did not subsequently realign with the mayor's political bloc. What changes is their political trajectory. External incoming managers ran for local office at roughly twice the rate of stayers, and that response increased with the size of the quota shock. The evidence is therefore more consistent with the managerial post becoming a platform for local political careers than with the direct appointment of partisan loyalists.

These findings qualify a common intuition about centralized social policy. Centralized rules can limit local discretion over eligibility and benefit levels, but they do not by themselves remove local politics from program delivery. When the last mile of implementation depends on municipal administrative effort, local incumbents can still be rewarded for converting federal authorization into visible benefits. The electoral return is not automatic: it appears where the local apparatus exists and where the personnel margin allows the apparatus to be used. In the language of the framework, federal policy creates the opportunity, but local infrastructure and front-line alignment determine whether that opportunity becomes realized enrollment.

More broadly, the results speak to attribution and state capacity. Voters can reward local incumbents for a federally funded program when the local component of implementation is visible, and the relevant capacity is not only bureaucratic quality but also the existence of the delivery apparatus and the political trajectory of those who operate it. Centralizing a transfer can discipline rules and payment flows, but it cannot fully centralize implementation politics when access still passes through local offices. Program design travels only as far as local capacity carries it.

References

Abadie, A., Athey, S., Imbens, G. W., and Wooldridge, J. M. (2023). When should you adjust standard errors for clustering? *Quarterly Journal of Economics*, 138(1):1–35.

Akhtari, M., Moreira, D., and Trucco, L. (2022). Political turnover, bureaucratic turnover, and the quality of public services. *American Economic Review*, 112(2):442–493.

Anderson, T. W. and Rubin, H. (1949). Estimation of the parameters of a single equation in a complete system of stochastic equations. *Annals of Mathematical Statistics*, 20(1):46–63.

Angrist, J. D. and Imbens, G. W. (1995). Two-stage least squares estimation of average causal effects in models with variable treatment intensity. *Journal of the American Statistical Association*, 90(430):431–442.

Ashworth, S. (2012). Electoral accountability: Recent theoretical and empirical work. *Annual Review of Political Science*, 15:183–201.

Avis, E., Ferraz, C., and Finan, F. (2018). Do government audits reduce corruption? Estimating the impacts of exposing corrupt politicians. *Journal of Political Economy*, 126(5):1912–1964.

Besley, T. (2006). *Principled Agents? The Political Economy of Good Government*. Oxford University Press, Oxford.

Besley, T. and Persson, T. (2009). The origins of state capacity: Property rights, taxation, and politics. *American Economic Review*, 99(4):1218–1244.

Besley, T. and Persson, T. (2011). *Pillars of Prosperity: The Political Economics of Development Clusters*. Princeton University Press, Princeton, NJ.

Blandhol, C., Bonney, J., Mogstad, M., and Torgovitsky, A. (2022). When is TSLS actually LATE? *NBER Working Paper*, (29709).

Borusyak, K. and Hull, P. (2023). Nonrandom exposure to exogenous shocks: Theory and applications. *Econometrica*, 91(6):2155–2185.

Brollo, F., Kaufmann, K., and La Ferrara, E. (2020). The political economy of program enforcement: Evidence from Brazil. *Journal of the European Economic Association*, 18(2):750–791.

Callaway, B., Goodman-Bacon, A., and Sant'Anna, P. H. C. (2024). Difference-in-differences with a continuous treatment. *NBER Working Paper No. 32117*. Working paper; see also Angrist & Imbens (1995) for the average-causal-response (ACR) framework.

Cameron, A. C., Gelbach, J. B., and Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427.

Colonnelli, E., Prem, M., and Teso, E. (2020). Patronage and selection in public sector organizations. *American Economic Review*, 110(10):3071–3099.

Dal Bó, E., Finan, F., and Rossi, M. A. (2013). Strengthening state capabilities: The role of financial incentives in the call to public service. *Quarterly Journal of Economics*, 128(3):1169–1218.

de Chaisemartin, C. (2017). Tolerating defiance? Local average treatment effects without monotonicity. *Quantitative Economics*, 8(2):367–396.

de Chaisemartin, C., D'Haultfœuille, X., and Vazquez-Bare, G. (2024). Difference-in-differences estimators with continuous treatments and no stayers. *arXiv preprint*, (2402.05432).

De La O, A. L. (2013). Do conditional cash transfers affect electoral behavior? Evidence from a randomized experiment in Mexico. *American Journal of Political Science*, 57(1):1–14.

Ferejohn, J. (1986). Incumbent performance and electoral control. *Public Choice*, 50(1–3):5–25.

Ferraz, C. and Finan, F. (2008). Exposing corrupt politicians: The effects of Brazil's publicly released audits on electoral outcomes. *Quarterly Journal of Economics*, 123(2):703–745.

Finan, F., Olken, B. A., and Pande, R. (2017). The personnel economics of the developing state. In Banerjee, A. V. and Duflo, E., editors, *Handbook of Economic Field Experiments*, volume 2, pages 467–514. North-Holland.

Fiszbein, A. and Schady, N. (2009). *Conditional Cash Transfers: Reducing Present and Future Poverty*. World Bank Policy Research Report. World Bank, Washington, DC.

Frey, A. (2019). Cash transfers, clientelism, and political enfranchisement: Evidence from Brazil. *Journal of Public Economics*, 176:1–17.

Fried, B. J. (2012). Distributive politics and conditional cash transfers: The case of Brazil's Bolsa Família. *World Development*, 40(5):1042–1053.

Gerard, F., Naritomi, J., and Silva, J. (2025). Cash transfers and the local economy: Evidence from Brazil. Working paper, 2025. Circulating identifiers: CEPR Discussion Paper 16286, World Bank Policy Research Working Paper 9778, and SSRN 3886759. Earlier version contains the technical appendix on the quota allocation formula.

Goldsmith-Pinkham, P., Hull, P., and Kolesár, M. (2022). Contamination bias in linear regressions. *NBER Working Paper*, (30108).

Hellmann, A. G. (2015). How does Bolsa Família work? Best practices in the implementation of conditional cash transfer programs in Latin America and the Caribbean. IDB Technical Note IDB-TN-856, Inter-American Development Bank.

Holmström, B. (1999). Managerial incentive problems: A dynamic perspective. *Review of Economic Studies*, 66(1):169–182.

Imai, K. and Kim, I. S. (2019). When should we use unit fixed effects regression models for causal inference with longitudinal data? *American Journal of Political Science*, 63(2):467–490.

Imbens, G. W. and Angrist, J. D. (1994). Identification and estimation of local average treatment effects. *Econometrica*, 62(2):467–475.

Jones, C. I. (2011). Intermediate goods and weak links in the theory of economic development. *American Economic Journal: Macroeconomics*, 3(2):1–28.

Klašnja, M. and Titiunik, R. (2017). The incumbency curse: Weak parties, term limits, and unfulfilled accountability. *American Political Science Review*, 111(1):129–148.

Kolesár, M. (2013). Estimation in an instrumental variables model with treatment effect heterogeneity. *Working Paper*. Princeton University.

Kremer, M. (1993). The O-ring theory of economic development. *Quarterly Journal of Economics*, 108(3):551–575.

Lindbeck, A. and Weibull, J. W. (1987). Balanced-budget redistribution as the outcome of political competition. *Public Choice*, 52(3):273–297.

Lindert, K., Linder, A., Hobbs, J., and de la Brière, B. (2007). The nuts and bolts of Brazil's Bolsa Família program: Implementing conditional cash transfers in a decentralized context. Social Protection Discussion Paper 0709, World Bank.

MacKinnon, J. G., Nielsen, M. Ø., and Webb, M. D. (2023). Cluster-robust inference: A guide to empirical practice. *Journal of Econometrics*, 232(2):272–299.

Manacorda, M., Miguel, E., and Vigorito, A. (2011). Government transfers and political support. *American Economic Journal: Applied Economics*, 3(3):1–28.

Maskin, E. and Tirole, J. (2004). The politician and the judge: Accountability in government. *American Economic Review*, 94(4):1034–1054.

McKenzie, D. (2012). Beyond baseline and follow-up: The case for more t in experiments. *Journal of Development Economics*, 99(2):210–221.

Nickell, S. (1981). Biases in dynamic models with fixed effects. *Econometrica*, 49(6):1417–1426.

Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics*, 37(2):187–204.

Pereira, T. S., Lucas, V. M., and Resende Filho, M. d. A. (2020). Ciclos políticos e eleitorais na execução do programa Bolsa Família em nível municipal. *Pesquisa e Planejamento Econômico (PPE)*, 50(1):67–98.

Przeworski, A., Stokes, S. C., and Manin, B., editors (1999). *Democracy, Accountability, and Representation*. Cambridge Studies in the Theory of Democracy. Cambridge University Press, Cambridge.

Robinson, J. A. and Verdier, T. (2013). The political economy of clientelism. *Scandinavian Journal of Economics*, 115(2):260–291.

Słoczyński, T. (2022). Interpreting ols estimands when treatment effects are heterogeneous: Smaller groups get larger weights. *Review of Economics and Statistics*, 104(3):501–509.

Sugiyama, N. B. and Hunter, W. (2013). Whither clientelism? Good governance and Brazil's Bolsa Família program. *Comparative Politics*, 46(1):43–62.

Zucco, C. (2013). When payouts pay off: Conditional cash transfers and voting behavior in Brazil 2002–10. *American Journal of Political Science*, 57(4):810–822.